16. *Assertion (A) :* Nuclei of different atoms have different sizes.

Reason (R) : The density of nuclear matter is same for all nuclei. *1*

CBSE 2026C

16. *Assertion (A) :* Nuclei of different atoms have different sizes.

Reason (R) : The density of nuclear matter is same for all nuclei. *1*

CBSE 2026C

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

27. Define binding energy of a nucleus. Differentiate between nucleus fusion and nuclear fission, giving one example of each. *3*

CBSE 2026C

27. Define binding energy of a nucleus. Differentiate between nucleus fusion and nuclear fission, giving one example of each. *3*

CBSE 2026C

If a certain number of neutrons and protons are brought together to form a nucleus of certain charge and mass, the energy (E_b) released in the process is called binding energy.

Nuclear Fusion	Nuclear Fission
A process in which **two or more lighter nuclei** ($A \leq 10$)join to form a **heavier nucleus** and energy is released. **Examples (Any one):** $^{1}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{2}_{1}\mathrm{H} + e^{+} + \nu + 0.42\ \mathrm{MeV}$ OR $^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{3}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n} + 3.27\ \mathrm{MeV}$ OR $^{2}_{1}\mathrm{H} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{3}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} + 4.03\ \mathrm{MeV}$	A process in which a **very heavy nucleus** breaks into **smaller nuclei** and energy is released. **Examples (Any one):** $^{1}_{0}\mathrm{n} + {}^{235}_{92}\mathrm{U} \rightarrow {}^{236}_{92}\mathrm{U} \rightarrow {}^{144}_{56}\mathrm{Ba} + {}^{89}_{36}\mathrm{Kr} + 3{}^{1}_{0}\mathrm{n}$ OR $^{1}_{0}\mathrm{n} + {}^{235}_{92}\mathrm{U} \rightarrow {}^{236}_{92}\mathrm{U} \rightarrow {}^{133}_{51}\mathrm{Sb} + {}^{99}_{41}\mathrm{Nb} + 4{}^{1}_{0}\mathrm{n}$ OR $^{1}_{0}\mathrm{n} + {}^{235}_{92}\mathrm{U} \rightarrow {}^{140}_{54}\mathrm{Xe} + {}^{94}_{38}\mathrm{Sr} + 2{}^{1}_{0}\mathrm{n}$

15. *Assertion (A) :* A free neutron is not a stable particle.

Reason (R) : It decays into a proton and an electron.

CBSE 2026C

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Reason (R) : It decays into a proton and an electron.

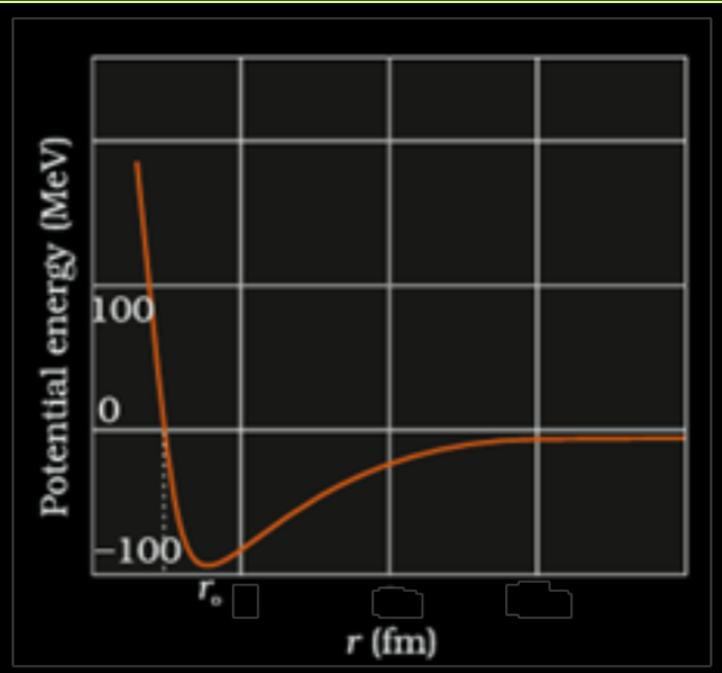
CBSE 2026C

(C) Assertion (A) is true, but Reason (R) is false.

20. Draw a plot showing the variation of potential energy of a pair of nucleons as a function of their separation. With the help of the plot, explain why the nuclear force is attractive for $r > r_0$ and repulsive for $r < r_0$. CBSE 2026C *2*

20. Draw a plot showing the variation of potential energy of a pair of nucleons as a function of their separation. With the help of the plot, explain why the nuclear force is attractive for $r > r_0$ and repulsive for $r < r_0$. 2

CBSE 2026C



1

If $r > r_0$, the potential energy gradually decreases with increasing r which indicates attractive force. ½

If $r < r_0$, the potential energy increases rapidly with decreasing r which indicates repulsive force. ½

8. Nuclides with the same number of neutrons are called :

(A) Isobars

(B) Isotones

(C) Isotopes

(D) Isomers

CBSE 2026

8. Nuclides with the same number of neutrons are called :

(A) Isobars

(B) Isotones

(C) Isotopes

(D) Isomers

CBSE 2026

(B) Isotone

9. The radius of a nucleus of mass number 125 is

(A) 6·0 fm

(B) 30 fm

(C) 72 fm

(D) 150 fm

CBSE 2026

9. The radius of a nucleus of mass number 125 is

(A) 6·0 fm

(B) 30 fm

(C) 72 fm

(D) 150 fm

CBSE 2026

(A) 6.0 fm(**Note. Award full marks for Attempt**)

15. *Assertion (A) :* Energy is released when heavy nuclei undergo fission or light nuclei undergo fusion.

Reason (R) : For heavy nuclei, binding energy per nucleon increases with increasing Z while for light nuclei, it decreases with increasing Z.

CBSE 2026

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Reason (R) : For heavy nuclei, binding energy per nucleon increases with increasing Z while for light nuclei, it decreases with increasing Z.

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(C) Assertion(A) is true, but reason(R) is false.

26. (a) (i) Write any two features of nuclear forces.

(ii) If both the number of protons and the neutrons are conserved in each nuclear reaction, in what way is mass converted into energy (or vice versa) in a nuclear reaction ? Explain. 3

CBSE 2026

26. (a) (i) Write any two features of nuclear forces.

(ii) If both the number of protons and the neutrons are conserved in each nuclear reaction, in what way is mass converted into energy (or vice versa) in a nuclear reaction ? Explain. CBSE 2026 *3*

(i) Features of Nuclear force: 1) charge independent 2) strongest force. 3) short-range force /saturated force 4) does not follow inverse square law. **Note: Award full marks for any two valid points.**	1+1
(ii) The total mass of the nuclei on the left-hand side need not be same as that on right hand side of nuclear reaction. Since Binding energy contributes to mass, the difference in these binding energies appear as energy released or absorbed in a nuclear reaction.	1

27. In the nuclear reaction :

$$_{1}^{2}\mathrm{H} + {}_{1}^{2}\mathrm{H} \rightarrow {}_{2}^{A}\mathrm{X} + {}_{0}^{1}\mathrm{n}$$

(a) Find the value of A.

(b) Calculate the amount of energy released in the reaction. *3*

Given : $m\left({}_{1}^{2}\mathrm{H}\right) = 2{\cdot}014102$ u, $m\left({}_{2}^{A}\mathrm{X}\right) = 3{\cdot}016049$ u

$m\left({}_{0}^{1}\mathrm{n}\right) = 1{\cdot}008665$ u, 1 u = 931·5 MeV/c^2

CBSE 2026

27. In the nuclear reaction :

$$ {}_{1}^{2}\mathrm{H} + {}_{1}^{2}\mathrm{H} \rightarrow {}_{2}^{A}\mathrm{X} + {}_{0}^{1}\mathrm{n} $$

(a) Find the value of A.

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CBSE 2026

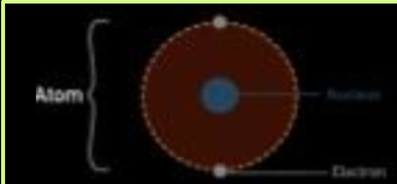
(a) ${}_{1}^{2}\mathrm{H} + {}_{1}^{2}\mathrm{H} \rightarrow {}_{2}^{A}\mathrm{X} + {}_{0}^{1}n$	
Mass number of reactants = Mass number of products	
$\therefore A = 3$	1
(b) Mass defect, ΔM = Mass of reactants − Mass of products	
$= \left[2 \times m({}_{1}^{2}\mathrm{H})\right] - \left[m({}_{2}^{A}\mathrm{X}) + m({}_{0}^{1}\mathrm{n})\right]$	½
$= [2 \times 2.014102] - [3.016049 + 1.0086654]$	
$\Delta M = 0.00349$ u	½
Energy released = $\Delta M \times 931.5\ \mathrm{MeV}/c^2$	½
$= 0.00349 \times 931.5$	
Energy released = 3.25 MeV	½

20. Prove that the density of nuclear matter is same for all nuclei. *2*

CBSE 2026

20. Prove that the density of nuclear matter is same for all nuclei. *2*

CBSE 2026



$$\text{Volume of a nucleus} = \frac{4}{3}\pi R^3$$

$$= \frac{4}{3}\pi(R_0 A^{\frac{1}{3}})^3 \quad \text{½}$$

$$= \frac{4}{3}\pi R_0^{\,3} A \quad \text{½}$$

$$\rho = \frac{\text{mass of nucleus}}{\text{volume of nucleus}} \quad \text{½}$$

$$= \frac{mA}{\frac{4}{3}\pi R_0^{\,3} A} = \frac{3m}{4\pi R_0^{\,3}} \quad \text{½}$$

21. Differentiate between nuclear fission and nuclear fusion, giving one example for each. *2*

CBSE 2026

21. Differentiate between nuclear fission and nuclear fusion, giving one example for each. *2*

CBSE 2026

Nuclear fission: When heavy nucleus splits into smaller nuclei.	½
Nuclear fusion: If two light nuclei combine to form a single larger nucleus.	½
Example (Nuclear fission): $^{1}_{0}n + {}^{235}_{92}U \rightarrow {}^{236}_{92}U \rightarrow {}^{144}_{56}Ba + {}^{89}_{36}Kr + 3\,{}^{1}_{0}n$	½
Example (Nuclear fusion): $^{1}_{1}H + {}^{1}_{1}H \rightarrow {}^{2}_{1}H + e^{+} + \upsilon + 0.42Mev$	½

10. If r_1 and r_2 are the radii of atomic nuclei of mass numbers 64 and 27 respectively, then the value of $\left(\frac{r_1}{r_2}\right)$ is :

(A) 1

(B) $\frac{4}{3}$

(C) $\frac{3}{4}$

(D) $\frac{27}{64}$

CBSE 2026

10. If r_1 and r_2 are the radii of atomic nuclei of mass numbers 64 and 27 respectively, then the value of $\left(\frac{r_1}{r_2}\right)$ is :

(A) 1

(B) $\frac{4}{3}$

(C) $\frac{3}{4}$

(D) $\frac{27}{64}$

CBSE 2026

(B) $\frac{4}{3}$

16. *Assertion (A) :* Nuclear forces are always attractive.

Reason (R) : The nuclear force between protons and neutrons in a nucleus is a weak force.

CBSE 2026

16. *Assertion (A) :* Nuclear forces are always attractive.

Reason (R) : The nuclear force between protons and neutrons in a nucleus is a weak force.

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(D) Both Assertion (A) and Reason (R) are false

19. Explain the terms mass defect and binding energy. How are they related ? *2*

CBSE 2026

19. Explain the terms mass defect and binding energy. How are they related ? 2

CBSE 2026

The difference in mass of a nucleus and its constituents is called mass defect.	½
If a certain number of neutrons and protons are brought together to form a nucleus of a certain mass and charge, the energy released is called binding energy.	½
$E_b = \Delta M c^2$	½
$= \lfloor [Zm_p + (A-Z)m_n] - M \rfloor c^2$	½

27. Differentiate between nuclear fission and nuclear fusion. Give one example for each with nuclear reaction. *3*

CBSE 2026

27. Differentiate between nuclear fission and nuclear fusion. Give one example for each with nuclear reaction. *3*

CBSE 2026

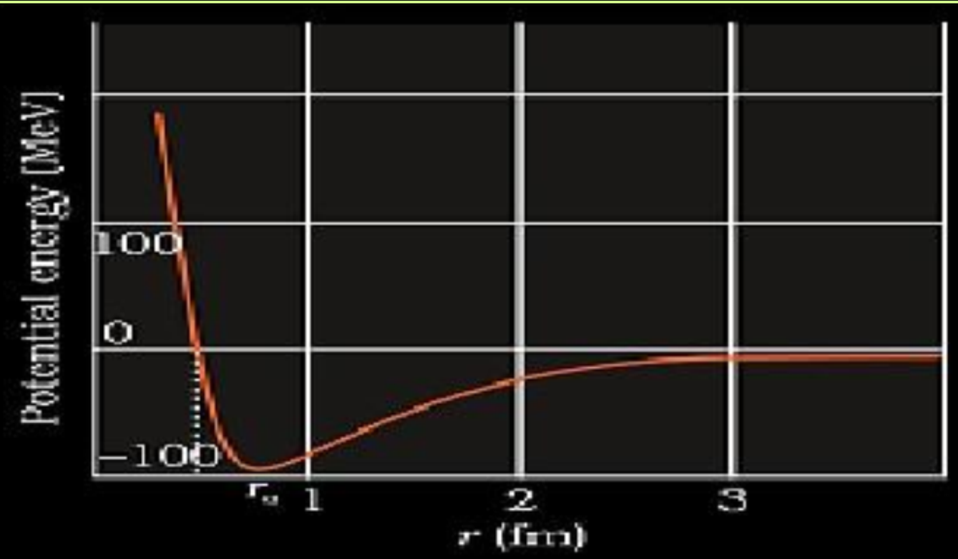
Nuclear fission is decay of heavy nucleus into two or more intermediate mass fragments whereas in nuclear fusion, light nuclei fuse into a heavier nucleus.	1
Nuclear Fission: $^{1}_{0}\text{n} + {}^{235}_{92}\text{U} \rightarrow {}^{144}_{56}\text{Ba} + {}^{89}_{36}\text{Kr} + 3{}^{1}_{0}\text{n}$	1
Nuclear Fusion: $^{1}_{1}\text{H} + {}^{1}_{1}\text{H} \rightarrow {}^{2}_{1}H + e^{+} + \nu + Q$	1

28. Draw a plot showing the variation of potential energy of a pair of nucleons as a function of their separation (r). Using this plot, show that the nuclear force is (i) attractive for $r > r_0$, and (ii) repulsive for $r < r_0$. *3*

CBSE 2026

28. Draw a plot showing the variation of potential energy of a pair of nucleons as a function of their separation (r). Using this plot, show that the nuclear force is (i) attractive for $r > r_0$, and (ii) repulsive for $r < r_0$. *3*

CBSE 2026



From the above plot, potential energy is minimum at distance $\mathbf{r_0}$. Therefore,	1
(i) the force is attractive for $r > r_0$.	½
(ii) the force is repulsive for $r < r_0$	½
(**Note**: Give full credit for this part if student depicts attractive and repulsive force in the graph)	

8. Which of the following particles do not exist in $_{92}U^{238}$ nucleus ? **1**

(A) 92 protons
(B) 92 electrons
(C) 146 neutrons
(D) 238 nucleons

CBSE 2026

(B) 92 electrons

9. The main source for energy generation in the stars is – **1**

(A) fusion of lighter nuclei (B) fission of heavy nuclei

(C) chemical reaction (D) fusion of heavy nuclei

CBSE 2026

9. The main source for energy generation in the stars is – **1**

(A) fusion of lighter nuclei (B) fission of heavy nuclei

(C) chemical reaction (D) fusion of heavy nuclei

CBSE 2026

(A) fusion of the lighter nuclei

22. (a) Consider the following nuclides : **3**

$^{12}_{6}C$, $^{198}_{80}Hg$, $^{14}_{6}C$, $^{197}_{79}Au$

Group them into isotopes and isotones.

(b) How does the size of a nucleus depend on its mass number A ? Hence prove that the density of nucleus is a constant, independent of A, for all nuclei.

CBSE 2026

22. (a) Consider the following nuclides : **3**

$^{12}_{6}C$, $^{198}_{80}Hg$, $^{14}_{6}C$, $^{197}_{79}Au$

Group them into isotopes and isotones.

(b) How does the size of a nucleus depend on its mass number A ? Hence prove that the density of nucleus is a constant, independent of A, for all nuclei.

CBSE 2026

(a) • Isotopes: $^{12}_{6}C$, $^{14}_{6}C$	$1/2$
• Isotones: $^{198}_{80}Hg$, $^{197}_{79}Au$	$1/2$
(b) • Size of nucleus α $A^{1/3}$	$1/2$
Alternatively:	
$R = R_0 A^{1/3}$	
• Suppose m is the average mass of a nucleon, density of nucleus	
$\rho = \frac{\text{Mass}}{\text{Volume}}$	$1/2$
$\rho = \frac{mA}{\frac{4}{3}\pi\left(R_0 A^{1/3}\right)^3}$	$1/2$
$\rho = \frac{3m}{4\pi R_0^3}$ = constant	$1/2$

15. **Reason (A) :** The mass of a nucleus is less than the sum of the masses of the constituent nucleons. **1**

Reason (R) : Energy is absorbed when the nucleons are bound together to form a nucleus.

CBSE 2026

15. **Reason (A) :** The mass of a nucleus is less than the sum of the masses of the constituent nucleons. **1**

Reason (R) : Energy is absorbed when the nucleons are bound together to form a nucleus.

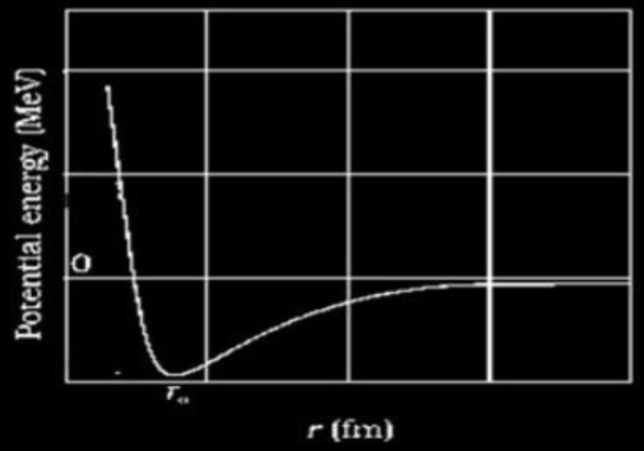
CBSE 2026

(C) Assertion (A) is true, but reason (R) is false.

21. Draw the plot of potential energy of a pair of nucleons as a function of their separation. Write two important conclusions that can be drawn from this plot. CBSE 2026 **2**

21. Draw the plot of potential energy of a pair of nucleons as a function of their separation. Write two important conclusions that can be drawn from this plot. CBSE 2026 **2**

- Graph



1

- Important conclusions

(i) Potential energy is minimum at a distance r_0 of about 0.8 fm

(ii) Force is attractive for distances larger than 0.8 fm.

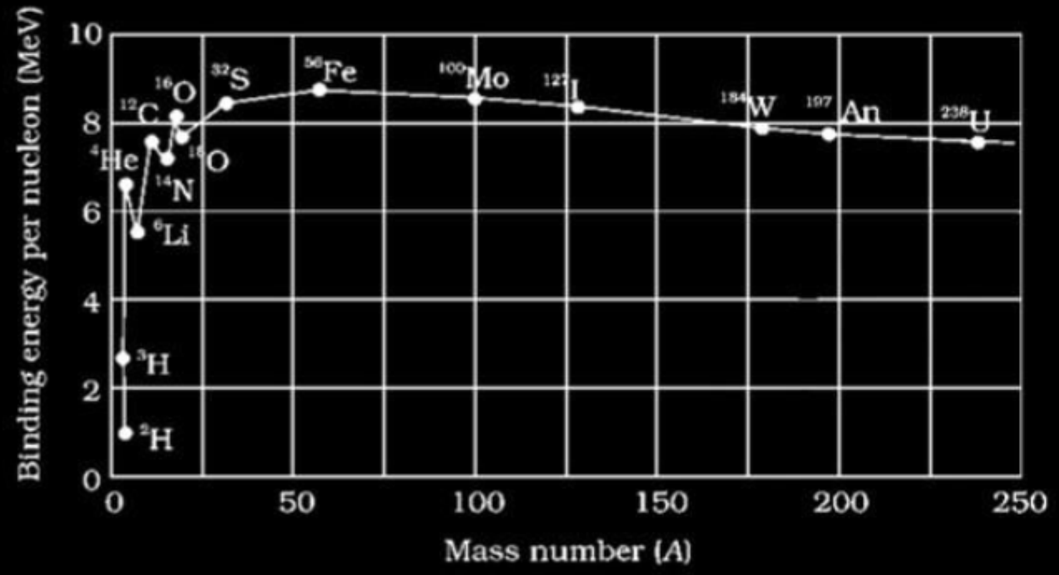
(iii) Force is repulsive for distances smaller than 0.8 fm

[Any two conclusions]

½+½

21. Draw a graph showing variation of binding energy per nucleon as a function of mass number A. The binding energy per nucleon for heavy nuclei (A > 170) decreases with increase in mass number. Explain its significance. CBSE 2026 **2**

21. Draw a graph showing variation of binding energy per nucleon as a function of mass number A. The binding energy per nucleon for heavy nuclei (A > 170) decreases with increase in mass number. Explain its significance. CBSE 2026 **2**



Note: [Give full marks for shape of graph only]

• For A>170, it signifies lesser stability

1½

½

21. Suppose a nucleus with mass number A = 240 and $\frac{\text{B.E.}}{\text{A}}$ = 7.6 MeV, breaks into two nuclei, each of mass number A = 120 with $\frac{\text{B.E.}}{\text{A}}$ = 8.5 MeV. Calculate the energy released in the process. CBSE 2026 **2**

21. Suppose a nucleus with mass number A = 240 and $\frac{\text{B.E.}}{\text{A}}$ = 7.6 MeV, breaks into two nuclei, each of mass number A = 120 with $\frac{\text{B.E.}}{\text{A}}$ = 8.5 MeV. Calculate the energy released in the process. CBSE 2026 **2**

E_{bn} for (A=240) Nucleus = 7.6 MeV	
E_{bn} for two fragments (A=120)	½
= 8.5 MeV	
Gain in E_{bn} = 8.5 – 7.6	½
= 0.9 MeV	½
Total gain in binding energy = 240 × 0.9	½
= 216MeV	
Alternatively:	
Binding energy for (A=240) nucleus is = 240 × 7.6 MeV	
Binding energy for fragments is = 2 × 120 × 8.5	
= 240 × 8.5 MeV	
Energy released = 240 × 8.5 – 240 × 7.6	
= 240 × (8.5 – 7.6)	
= 240 × 0.9	
= 216 MeV	

10. The potential energy of a pair of nucleons is minimum when they are separated by about :

(A) $0{\cdot}8 \times 10^{-15}$ m

(B) $0{\cdot}8 \times 10^{-10}$ m

(C) $2{\cdot}0 \times 10^{-13}$ m

(D) $2{\cdot}0 \times 10^{-14}$ m

CBSE 2025 C

10. The potential energy of a pair of nucleons is minimum when they are separated by about :

(A) $0{\cdot}8 \times 10^{-15}$ m

(B) $0{\cdot}8 \times 10^{-10}$ m

(C) $2{\cdot}0 \times 10^{-13}$ m

(D) $2{\cdot}0 \times 10^{-14}$ m

CBSE 2025 C

(A) 0.8×10^{-15} m

24. Define binding energy (BE) of a nucleus. An unstable nucleus has mass number A = 240 and $\frac{BE}{A}$ = 7·6 MeV. It splits in two fragments, each of A = 120 with $\frac{BE}{A}$ = 8·5 MeV. Calculate the energy released. CBSE 2025 C *3*

24. Define binding energy (BE) of a nucleus. An unstable nucleus has mass number A = 240 and $\frac{BE}{A}$ = 7·6 MeV. It splits in two fragments, each of A = 120 with $\frac{BE}{A}$ = 8·5 MeV. Calculate the energy released. CBSE 2025 C *3*

Binding energy : Energy released in bringing all the nucleons of a nucleus together to form a nucleus. Alternatively – Energy required to separate the nucleons from the nucleus.	1
Binding Energy of X, $B.E_X = 240 \times 7.6$ $= 1824$ MeV	½
Binding Energy of Y, $B.E_Y = 120 \times 8.5 = 1020$ MeV	½
Energy released $E = 2(B.E_Y) - (B.E_X)$	½
$= (2 \times 1020) - 1824$	½
$= 216$ MeV	

26. What is thermonuclear fusion ?

Calculate the energy released in MeV in the fusion reaction given below :

$$_{1}^{2}H + {}_{1}^{3}H \longrightarrow {}_{2}^{4}He + n$$

Given : $m(_{1}^{2}H) = 2{\cdot}014102$ u, $m(_{1}^{3}H) = 3{\cdot}016049$ u, $m(_{2}^{4}He) = 4{\cdot}002603$ u,

$m_n = 1{\cdot}008665$ u, 1 u = 930 MeV/c^2 CBSE 2025 C *3*

26. What is thermonuclear fusion ?

Calculate the energy released in MeV in the fusion reaction given below :

$$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \longrightarrow {}^{4}_{2}\mathrm{He} + \mathrm{n}$$

Given : $m(^{2}_{1}\mathrm{H}) = 2{\cdot}014102$ u, $m(^{3}_{1}\mathrm{H}) = 3{\cdot}016049$ u, $m(^{4}_{2}\mathrm{He}) = 4{\cdot}002603$ u,

$m_n = 1{\cdot}008665$ u, 1 u = 930 MeV/c^2 CBSE 2025 C *3*

Thermonuclear fusion:	1
When two light nuclei fuse to form a larger nucleus, energy is released	
Mass defect	½
$\Delta m = \left[m\left(^{2}_{1}H\right) + m\left(^{3}_{1}H\right)\right] - \left[m\left(^{4}_{2}He\right) + m_n\right]$	
$= (2.014102 + 3.016049) - (4.002603 + 1.008665)$	½
$\Delta m = 0.018883$ u	
Energy released	½
$E = \Delta m \times 930$ MeV	
$= 0.018883 \times 930$	½
$= 17.56$ MeV	

26. Give an example of fission reaction. Calculate the energy released by the fission of 1 kg of $^{235}_{92}U$ in kWh. The energy released per fission is 200 MeV. *3*

CBSE 2025 C

26. Give an example of fission reaction. Calculate the energy released by the fission of 1 kg of $^{235}_{92}U$ in kWh. The energy released per fission is 200 MeV. *3*

CBSE 2025 C

- Example of fission reaction.

$$^{1}_{0}n + {}^{235}_{92}U \rightarrow {}^{144}_{56}Ba + {}^{89}_{36}Kr + 3\,{}^{1}_{0}n$$ — 1

Or any other appropriate example of fission reaction.

- No. of atoms in 1 kg of $^{235}_{92}U = \frac{6.023 \times 10^{23}}{235} \times 1000$ — ½

Total energy released by the fission of 1 kg of $^{235}_{92}U$ — ½

$$= \frac{6.023 \times 10^{26}}{235} \times 200 \text{ MeV}$$

$$= \frac{6.023 \times 10^{26}}{235} \times 200 \times 1.6 \times 10^{-13} \text{ J}$$

$$= \frac{6.023 \times 10^{26} \times 200 \times 1.6 \times 10^{-13}}{235 \times 3.6 \times 10^{6}} \text{ kWh}$$ — ½

$$= 2.27 \times 10^{7} \text{ kWh}$$ — ½

10. If 1·0 milligram of mass is converted into energy, the energy released will be :

(A) 9×10^{16} J

(B) 3×10^{12} J

(C) 9×10^{10} J

(D) 3×10^{8} J

CBSE 2025 C

10. If 1·0 milligram of mass is converted into energy, the energy released will be :

(A) 9×10^{16} J

(B) 3×10^{12} J

(C) 9×10^{10} J

(D) 3×10^{8} J

CBSE 2025 C

(C) 9×10^{10} J

13. *Assertion (A) :* Nuclear forces are always attractive in nature.

Reason (R) : Nuclear forces are charge-dependent forces.

CBSE 2025 C

13. *Assertion (A) :* Nuclear forces are always attractive in nature.

Reason (R) : Nuclear forces are charge-dependent forces.

CBSE 2025 C

(D) Both Assertion (A) and Reason (R) are false.

(b) Find the ratio of nuclear radii of Iodine $\left({}^{125}_{53}\mathrm{I}\right)$ and Aluminium $\left({}^{27}_{13}\mathrm{Al}\right)$.

CBSE 2025 C

2

(b) Find the ratio of nuclear radii of Iodine $\left({}^{125}_{53}\mathrm{I}\right)$ and Aluminium $\left({}^{27}_{13}\mathrm{Al}\right)$.

CBSE 2025 C

2

$R = R_0 A^{1/3}$	½
$\frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{\frac{1}{3}}$	½
$\frac{R_1}{R_2} = \left(\frac{125}{27}\right)^{1/3}$	½
$\frac{R_1}{R_2} = \frac{5}{3}$	½

26. (a) Differentiate between isobars and isotones.

(b) Show that the nuclear density is independent of mass number A of a nucleus. CBSE 2025 C *3*

26. (a) Differentiate between isobars and isotones.

(b) Show that the nuclear density is independent of mass number A of a nucleus. CBSE 2025 C *3*

(a) Nucleides with same mass number are called isobars. Nucleides with same neutron number but different atomic numbers are called isotones. (b) If R is the radius of nucleus, then its volume $V = \frac{4}{3}\pi R^3$	1
As $R = R_o A^{\frac{1}{3}}$, $V = \frac{4}{3}\pi R_o^3 A$	½
Density of nucleus $\rho = \frac{M}{V} = \frac{A}{\frac{4}{3}\pi R_o{}^3 A}$	½
$\rho = \frac{3}{4\pi R_o^3} = \text{constant}$	½
The nuclear density is independent of A.	½

5. Isotones are the nuclides having :

(A) same mass numbers

(B) same atomic numbers

(C) same neutron number, but different atomic number

(D) different neutron number, and different mass number

CBSE 2025

5. Isotones are the nuclides having :

(A) same mass numbers

(B) same atomic numbers

(C) same neutron number, but different atomic number

(D) different neutron number, and different mass number

CBSE 2025

(C) same neutron number, but different atomic number.

10. The mass numbers of two nuclei A and B are 27 and 64 respectively. The ratio of their radii $\left(\frac{r_A}{r_B}\right)$ will be :

(A) $\frac{27}{64}$

(B) $\frac{9}{16}$

(C) $\frac{3\sqrt{3}}{8}$

(D) $\frac{3}{4}$

CBSE 2025

10. The mass numbers of two nuclei A and B are 27 and 64 respectively. The ratio of their radii $\left(\frac{r_A}{r_B}\right)$ will be :

(A) $\frac{27}{64}$

(B) $\frac{9}{16}$

(C) $\frac{3\sqrt{3}}{8}$

(D) $\frac{3}{4}$

CBSE 2025

(D) 3/4

21. Calculate the binding energy per nucleon (in MeV) of a helium nucleus (^4_2He). *2*

Given : $m(^4_2\text{He}) = 4{\cdot}002603$ u

$m_n = 1{\cdot}008665$ u

$m_H = 1{\cdot}007825$ u

1 u $= 931{\cdot}5$ MeV/c^2

CBSE 2025

21. Calculate the binding energy per nucleon (in MeV) of a helium nucleus (^4_2He). *2*

Given : $m(^4_2\text{He}) = 4{\cdot}002603$ u

$m_n = 1{\cdot}008665$ u

$m_H = 1{\cdot}007825$ u

$1\text{ u} = 931{\cdot}5\text{ MeV}/c^2$

CBSE 2025

$\Delta m = (2m_n + 2m_H) - m(^4_2He)$	½
$= 2\times1.008665 + 2\times1.007825 - 4.002603$	
$= 0.0303774u$	½
$BE = \Delta mc^2$	
$= 0.030377\times931$	
$= 28.2962\,\text{MeV}$	½
$\frac{\text{BE}}{\text{nucleon}} = \frac{28.2962}{4} = 7.07\,\text{MeV}$	½

21. Briefly explain how energy is produced in stars, giving two examples of the nuclear reactions involved. CBSE 2025 *2*

21. Briefly explain how energy is produced in stars, giving two examples of the nuclear reactions involved. CBSE 2025 *2*

Energy is produced in stars due to Nuclear fusion.	½
Two lighter nuclei fuse to form a heavier nucleus which is more stable, hence energy is released.	½
Alternatively: Due the difference in masses in reactants and products, mass is converted into energy. Hence energy is released.	
$^{1}_{1}H + {}^{1}_{1}H \rightarrow {}^{2}_{1}He + e^{+} + \nu$	
$^{2}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{2}He + {}^{1}_{0}n$	½+ ½
$^{3}_{2}He + {}^{3}_{2}He \rightarrow {}^{4}_{2}He + {}^{1}_{1}H + {}^{1}_{1}H$	
Any two above equation.	

5. Inside a nucleus, the nuclear forces between proton and proton, proton and neutron, neutron and neutron are F_{pp}, F_{pn} and F_{nn} respectively. Then :

(A) $F_{pp} > F_{pn} > F_{nn}$

(B) $F_{pn} > F_{nn} > F_{pp}$

(C) $F_{nn} > F_{pp} > F_{pn}$

(D) $F_{pp} = F_{pn} = F_{nn}$

CBSE 2025

5. Inside a nucleus, the nuclear forces between proton and proton, proton and neutron, neutron and neutron are F_{pp}, F_{pn} and F_{nn} respectively. Then :

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(B) $F_{pn} > F_{nn} > F_{pp}$

(C) $F_{nn} > F_{pp} > F_{pn}$

(D) $F_{pp} = F_{pn} = F_{nn}$

CBSE 2025

(D) $F_{pp} = F_{pn} = F_{nn}$

14. *Assertion (A) :* The binding energy per nucleon is practically constant for mass number in the range ($30 < A < 170$).

Reason (R) : Nuclear forces between the nucleons for mass numbers in the range ($30 < A < 170$) are not short-range. CBSE 2025

14. *Assertion (A) :* The binding energy per nucleon is practically constant for mass number in the range ($30 < A < 170$).

Reason (R) : Nuclear forces between the nucleons for mass numbers in the range ($30 < A < 170$) are not short-range. CBSE 2025

(C) Assertion (A) is true, But Reason (R) is false.

23. (a) Differentiate between 'nuclear fission' and 'nuclear fusion'. Briefly discuss one example of each.

(b) Draw a graph of potential energy between a pair of nucleons as a function of their separation. CBSE 2025 *3*

23. (a) Differentiate between 'nuclear fission' and 'nuclear fusion'. Briefly discuss one example of each.

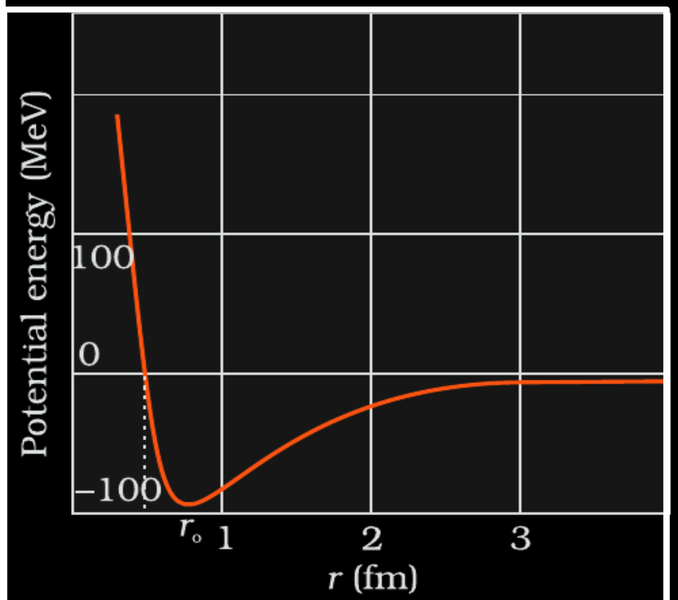
(b) Draw a graph of potential energy between a pair of nucleons as a function of their separation. CBSE 2025 *3*

(a) **Nuclear fission** is the process of splitting up of a heavy nucleus into lighter ones with a release of energy. ½

$${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{236}_{92}U \rightarrow {}^{144}_{56}Ba + {}^{89}_{36}Kr + 3\,{}^{1}_{0}n$$ (or any other reaction) ½

Nuclear fusion is the process of fusing of two lighter nuclei to form a heavier nucleus with the release of energy. ½

$${}^{1}_{1}H + {}^{1}_{1}H \rightarrow {}^{2}_{1}H + e^{+} + \nu + 0.42\,\text{MeV}$$ (or any other reaction) ½



1

26. Find the Q value of the following nuclear reaction :

$$^{12}_{6}C + ^{12}_{6}C \longrightarrow ^{20}_{10}Ne + ^{4}_{2}He$$

Is this reaction exothermic or endothermic ? *3*

Given :

$$m\left(^{12}_{6}C\right) = 12 \cdot 000000 \text{ u}$$

$$m\left(^{20}_{10}Ne\right) = 19 \cdot 992439 \text{ u}$$

$$m\left(^{4}_{2}He\right) = 4 \cdot 002603 \text{ u}$$

$$1 \text{ u} = 931 \text{ MeV/c}^2$$

CBSE 2025

26. Find the Q value of the following nuclear reaction :

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$$\mathrm{m}\left(^{12}_{6}\mathrm{C}\right) = 12{\cdot}000000\ \mathrm{u}$$

$$\mathrm{m}\left(^{20}_{10}\mathrm{Ne}\right) = 19{\cdot}992439\ \mathrm{u}$$

$$\mathrm{m}\left(^{4}_{2}\mathrm{He}\right) = 4{\cdot}002603\ \mathrm{u}$$

$1\ \mathrm{u} = 931\ \mathrm{MeV/c^2}$

CBSE 2025

Mass defect Δm = mass of the reactants – mass of the products	½
$\Delta m = 2\times12 - 19.992439 - 4.002603$	½
$\Delta m = 0.004958u$	½
$Q = \Delta m \times 931$	½
$= 0.004958 \times 931$	
$= 4.62\,\mathrm{MeV}$	½
The reaction is exothermic.	½

28. (a) Consider the so-called 'D-T reaction' (Deuterium-Tritium reaction). In a thermonuclear fusion reactor, the following nuclear reaction occurs :

$$_{1}^{2}\mathrm{H} + _{1}^{3}\mathrm{H} \longrightarrow _{2}^{4}\mathrm{He} + _{0}^{1}\mathrm{n} + \mathrm{Q}$$

Find the amount of energy released in the reaction. CBSE 2025

Given :

$m\left(_{1}^{2}\mathrm{H}\right) = 2{\cdot}014102 \text{ u}$

$m\left(_{1}^{3}\mathrm{H}\right) = 3{\cdot}016049 \text{ u}$

$m\left(_{2}^{4}\mathrm{He}\right) = 4{\cdot}002603 \text{ u}$

$m\left(_{0}^{1}\mathrm{n}\right) = 1{\cdot}008665 \text{ u}$

$1 \text{ u} = 931 \text{ MeV/c}^2$

(b) Show that the nuclear density is independent of mass number. *3*

$\Delta m = \left[m(^{2}_{1}H) + m(^{3}_{1}H) \right] - \left[m(^{4}_{2}He) + m(^{1}_{0}\text{n}) \right]$	½
$= (2.014102 + 3.016049) - (4.002603 + 1.008665)$	
$= 0.018883\text{u}$	½
$Q = \Delta m \times 931$	
$= 0.018883 \times 931\ \text{MeV}$	½
$Q = 17.58\ \text{MeV}$	½
$\text{Nuclear density} = \dfrac{\text{Mass of nucleus}}{\text{Volume of nucleus}}$	
$\rho = \dfrac{mA}{\frac{4}{3}\pi R^3}$	
$R = R_0 A^{1/3}$	½
$\rho = \dfrac{3m}{4\pi R_0^3}$	½
Independent of mass number (A)	

13. *Assertion (A) :* During formation of a nucleus, the mass defect produced is the source of the binding energy of the nucleus.

Reason (R) : For all nuclei, the value of binding energy per nucleon increases with mass number.

CBSE 2025

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Reason (R) : For all nuclei, the value of binding energy per nucleon increases with mass number.

CBSE 2025

(C) Assertion (A) is true, but Reason (R) is false

20. (a) Why is the mass of a nucleus always less than the sum of the masses of its constituents, i.e. free neutrons and free protons ?

(b) How is Coulomb repulsion between protons in a nucleus overcome ? Explain. CBSE 2025 *2*

20. (a) Why is the mass of a nucleus always less than the sum of the masses of its constituents, i.e. free neutrons and free protons ?

(b) How is Coulomb repulsion between protons in a nucleus overcome ? Explain.

CBSE 2025 *2*

(a) Because of binding of nucleus some energy is released and hence mass decreases during formation of nucleus.	**1**
(b) By very strong attraction due to nuclear forces	**1**

20. When a neutron collides with $^{235}_{92}U$, the nucleus gives $^{140}_{54}Xe$ and $^{94}_{38}Sr$ as fission products and two neutrons are ejected. Calculate the mass defect and the energy released (in MeV) in the process. Given : *2*

$m\left(^{235}_{92}U\right) = 235{\cdot}04393$ u, $m\left(^{140}_{54}Xe\right) = 139{\cdot}92164$ u

$m\left(^{94}_{38}Sr\right) = 93{\cdot}91536$ u, $^{1}_{0}n = 1{\cdot}00866$ u

1 u $= 931$ MeV/c^2

CBSE 2025

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$$m\left(^{94}_{38}\mathrm{Sr}\right) = 93{\cdot}91536\ \mathrm{u}, \quad ^{1}_{0}\mathrm{n} = 1{\cdot}00866\ \mathrm{u}$$

$$1\ \mathrm{u} = 931\ \mathrm{MeV}/c^2$$

CBSE 2025

$^{235}_{92}U + {}^{1}_{0}\mathrm{n} \rightarrow {}^{192}_{54}Xe + {}^{94}_{38}Sr + 2({}^{1}_{0}n)$	½
$\Delta m = m({}^{1}_{0}n) + m({}^{235}_{92}\mathrm{U}) - (m({}^{140}_{54}\mathrm{Xe}) + m({}^{94}_{38}\mathrm{Sr}) + 2\times m({}^{1}_{0}\mathrm{n}))$	½
$= 1.00866 + 235.04393 - 139.92164 - 93.91536 - 2\times 1.00866$	½
$= 0.19827\,u$	
Energy released $= \Delta m \times 931\mathrm{MeV}$	
$= 0.19827\times 931\ \mathrm{MeV}$	
$= 184.59\ \mathrm{MeV}$	½

20. Calculate the mass of an α-particle in atomic mass unit (u). Given,
Mass of a normal helium atom = 4·002603 u
Mass of carbon atom = $1{\cdot}9926 \times 10^{-26}$ kg CBSE 2025 *2*

20. Calculate the mass of an α-particle in atomic mass unit (u). Given,
Mass of a normal helium atom = 4·002603 u
Mass of carbon atom = $1{\cdot}9926 \times 10^{-26}$ kg CBSE 2025 *2*

$1\text{u} = \frac{1}{12}\text{mass of carbon atom} = \frac{1.9926\times10^{-26}\text{kg}}{12} = 1.66\times10^{-27}\,kg$	
$\text{mass of an electron} = 9.1\times10^{-31}\text{kg}$	½
$\text{mass of two electrons} = \frac{2\times9.1\times10^{-31}}{1.66\times10^{-27}}$	½
$= 0.00109638\text{ u}$	
mass of α-particle = mass of the normal helium atom - mass of two electrons	½
$= 4.0026030 - 0.00109638$	½
$= 4.00150662\text{ u}$	

23. (a) Show the variation of binding energy per nucleon with mass number. Write the significance of the binding energy curve. **3**

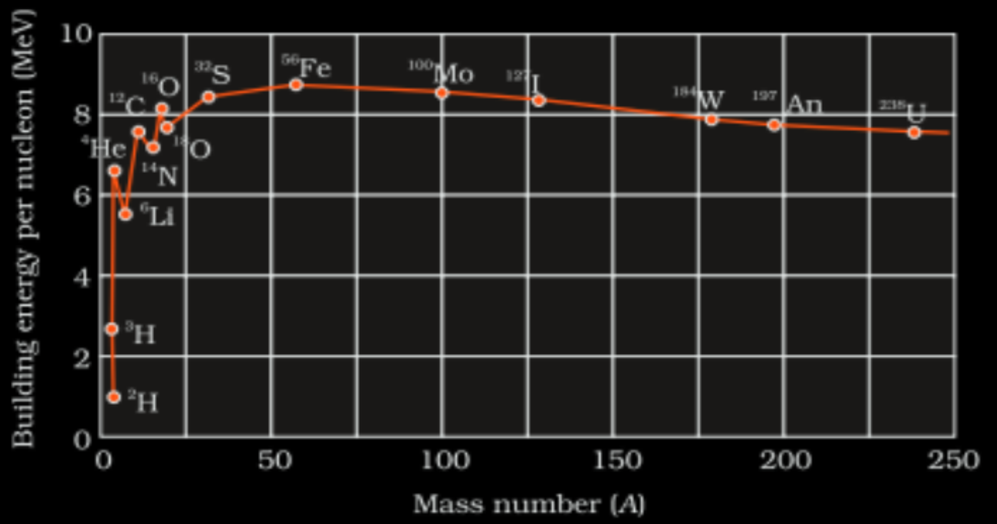
(b) Two nuclei with lower binding energy per nucleon form a nuclei with more binding energy per nucleon.

(i) What type of nuclear reaction is it ?

(ii) Whether the total mass of nuclei increases, decreases or remains unchanged ?

(iii) Does the process require energy or produce energy ? CBSE 2025

(a)



1

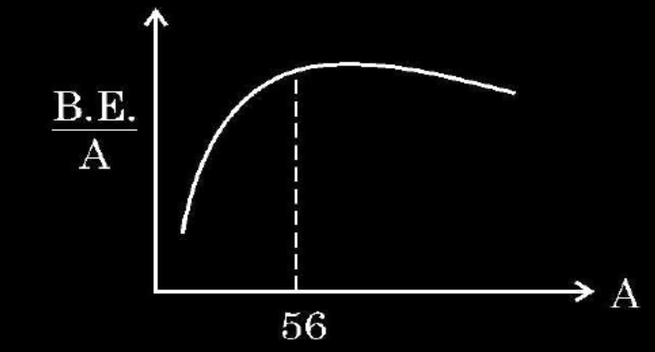
Note: - Full credit to be given even if the values are not shown.

Significance of the binding energy curve –
(Any one) ½
- Why lighter nuclei undergo fusion and heavier nuclei undergo fission.
- Nuclear forces are short ranged.
- Energy is released in both nuclear fission and nuclear fusion.

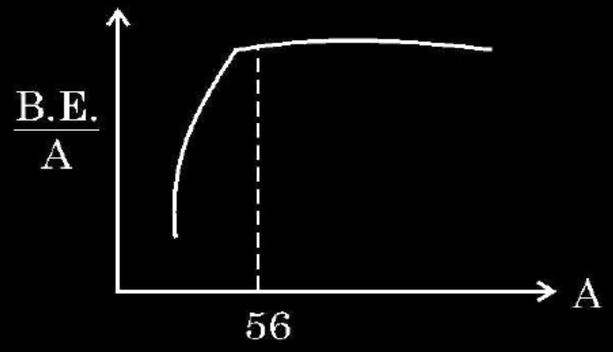
(b) (i) Nuclear fusion ½
(ii) Decreases ½
(iii) Energy is produced ½

12. Which of the following figures correctly represent the shape of curve of binding energy per nucleon as a function of mass number ? **1**

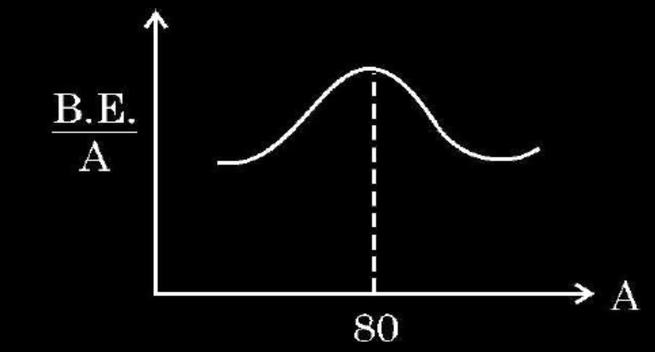
(A)



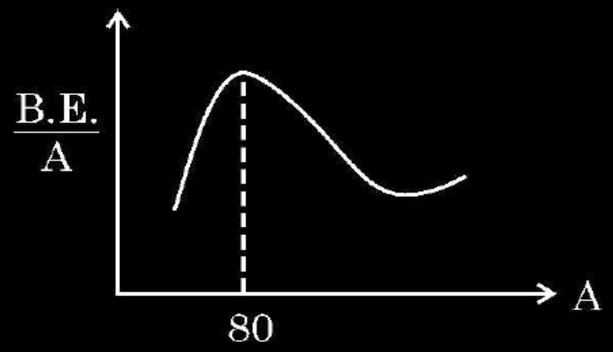
(B)



(C)



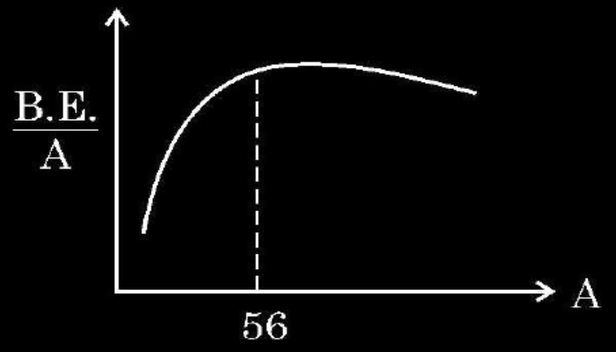
(D)



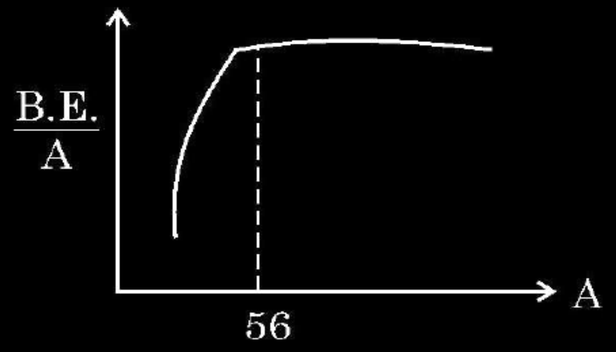
CBSE 2025

12. Which of the following figures correctly represent the shape of curve of binding energy per nucleon as a function of mass number ? **1**

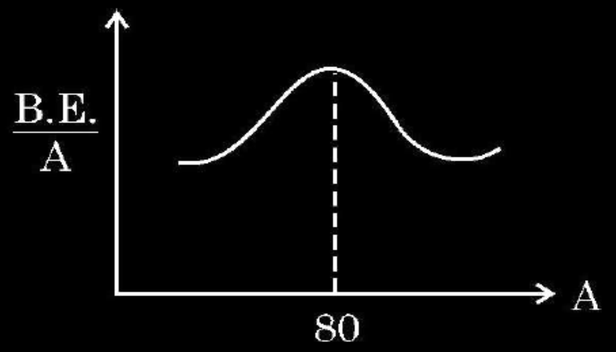
(A)



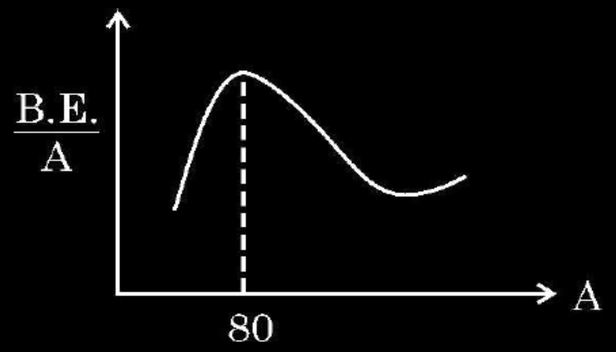
(B)



(C)

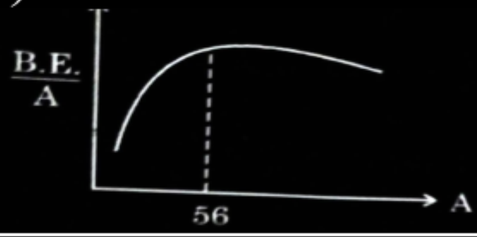


(D)



CBSE 2025

(A)



28. (a) Define 'Mass defect' and 'Binding energy' of a nucleus. Describe 'Fission process' on the basis of binding energy per nucleon. **3**

(b) A deuteron contains a proton and a neutron and has a mass of 2.013553 u. Calculate the mass defect for it in u and its energy equivalence in MeV. (m_p = 1.007277 u, m_n = 1.008665 u, 1u = 931.5 MeV/c^2)

CBSE 2024 C

28. (a) Define 'Mass defect' and 'Binding energy' of a nucleus. Describe 'Fission process' on the basis of binding energy per nucleon. **3**

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CBSE 2024 C

a) Difference in the mass of the nucleus and its constituents is defined as mass defect.	½
Binding Energy is the energy required to separate the nucleons from the nucleus.	½
In Fission process a heavy nucleus splits into lighter nuclei and energy is released. As a result the Binding Energy per nucleon increases.	½
b) $\Delta m = (m_p + m_n) - m_d$	½
$\Delta m = (1.007277 + 1.008665) - 2.013553$	
$\Delta m = 0.002389$ u	½
Energy released $= \Delta m \times c^2$	
Energy released $= 0.002389 \times 931.5$	
$= 2.2253$ MeV ≈ 2.22 MeV	½

3. m, m_n and m_p represents masses of ${}^{A}_{Z}X$ nucleus, a neutron and a proton, respectively. Then :

(A) $m < (A - Z)\, m_n + Z\, m_p$

(B) $m = (A - Z)\, m_n + Z\, m_p$

(C) $m = (A - Z)\, m_p + Z\, m_n$

(D) $m > (A - Z)\, m_n + Z\, m_p$

CBSE 2024 C

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(D) $m > (A - Z)\, m_n + Z\, m_p$

CBSE 2024 C

(A) $M < (A-Z)\, m_n + Z m_p$

15. *Assertion (A) :* Nuclear fission reactions are responsible for energy generation in the Sun.

Reason (R) : Light nuclei fuse together in the nuclear fission reactions.

CBSE 2024 C

15. *Assertion (A) :* Nuclear fission reactions are responsible for energy generation in the Sun.

Reason (R) : Light nuclei fuse together in the nuclear fission reactions.

CBSE 2024 C

(D) Both Assertion (A) and Reason (R) are false.

(ii) Calculate binding energy per nucleon (in MeV) of $^{12}_{6}C$.

Given :

$$m\left(^{12}_{6}C\right) = 12.000000 \text{ u}$$

$$m\left(^{1}_{0}n\right) = 1.008665 \text{ u}$$

$$m\left(^{1}_{1}H\right) = 1.007825 \text{ u}$$

CBSE 2024 C

(ii) Calculate binding energy per nucleon (in MeV) of $^{12}_{6}C$.

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$$m\left(^{1}_{1}H\right) = 1.007825 \text{ u}$$

CBSE 2024 C

(ii)

Mass defect, $\Delta m = \left[6m(^{1}_{0}n) + 6m(^{1}_{1}H)\right] - m(^{12}_{6}C)$ ½

$\Delta m = (6\times 1.008665 + 6\times 1.007825) - 12.000000$

$\Delta m = 0.09894\,\text{u}$ ½

$B.E. = \Delta m \times 931.5\,\text{MeV}$

$= 92.16\text{MeV}$

Binding energy per nucleon, $E_{bn} = \dfrac{E_b}{A}$

$= \dfrac{92.16}{12}$

$= 7.68\text{MeV}$ ½

5. Isotones are nuclides having :

(A) same number of neutrons but different number of protons

(B) same number of protons but different number of neutrons

(C) same number of protons and also same number of neutrons

(D) different number of protons and also different number of neutrons

CBSE 2024 C

5. Isotones are nuclides having :

(A) same number of neutrons but different number of protons

(B) same number of protons but different number of neutrons

(C) same number of protons and also same number of neutrons

(D) different number of protons and also different number of neutrons

CBSE 2024 C

(A) Same number of neutrons but different number of protons

6. The potential energy between two nucleons inside a nucleus is minimum at a distance of about

(A) 0.8 fm
(B) 1.6 fm
(C) 2.0 fm
(D) 2.8 fm

CBSE 2024

6. The potential energy between two nucleons inside a nucleus is minimum at a distance of about

(A) 0.8 fm
(B) 1.6 fm
(C) 2.0 fm
(D) 2.8 fm

CBSE 2024

(A) 0.8 fm

20. Deuterium undergoes the following fusion reaction : CBSE 2024

$$_{1}^{2}\mathrm{H} + {}_{1}^{2}\mathrm{H} \longrightarrow {}_{2}^{3}\mathrm{He} + {}_{0}^{1}\mathrm{n} + 3.27\ \mathrm{MeV}$$

How long an electric bulb of 200 W will glow by using the energy released in 2 g of deuterium ?

20. Deuterium undergoes the following fusion reaction : CBSE 2024

$$_{1}^{2}\mathrm{H} + _{1}^{2}\mathrm{H} \longrightarrow _{2}^{3}\mathrm{He} + _{0}^{1}\mathrm{n} + 3.27\ \mathrm{MeV}$$

How long an electric bulb of 200 W will glow by using the energy released in 2 g of deuterium ?

(a) Number of atoms in 2g deuterium $= 6.023\times10^{23}$	½
Energy released /atom $= \dfrac{3.27}{2} = 1.635\,\mathrm{MeV}$	½
$t = \dfrac{\text{Total energy released}}{\text{Power}}$	½
$t = \dfrac{6.023\times10^{23}\times1.635\times1.6\times10^{-13}}{200}$	
$t = 7.88\times10^{8}\ \mathrm{s}$	½

20. Calculate the energy released/absorbed (in MeV) in the nuclear reaction :

$$_{1}^{1}\mathrm{H} + {}_{1}^{3}\mathrm{H} \longrightarrow {}_{1}^{2}\mathrm{H} + {}_{1}^{2}\mathrm{H}$$

Given : $m\left({}_{1}^{1}\mathrm{H}\right) = 1.007825\ \mu$

$m\left({}_{1}^{2}\mathrm{H}\right) = 2.014102\ \mu$

$m\left({}_{1}^{3}\mathrm{H}\right) = 3.016049\ \mu$

CBSE 2024

20. Calculate the energy released/absorbed (in MeV) in the nuclear reaction :

$$_1^1\mathrm{H} + {}_1^3\mathrm{H} \longrightarrow {}_1^2\mathrm{H} + {}_1^2\mathrm{H}$$

Given : $m\left({}_1^1\mathrm{H}\right) = 1.007825\ \mu$

$m\left({}_1^2\mathrm{H}\right) = 2.014102\ \mu$

$m\left({}_1^3\mathrm{H}\right) = 3.016049\ \mu$

CBSE 2024

Mass of reactants = (1.007825 + 3.016049) u = 4.023874 u	
Mass of product = 2 × 2.014102 u = 4.028204 u	
Mass defect , Δm = 4.023874 u – 4.028204 u = - 0.00433 u	1
As the mass defect is negative, energy is absorbed.	½
Energy absorbed, E = 0.00433 × 931.5 MeV = 4.03 MeV	½

28. (a) Differentiate between nuclear fission and fusion. **3**

(b) The fission properties of $_{94}Pu^{239}$ are very similar to those of $_{92}U^{235}$. How much energy (in MeV), is released if all the atoms in 1 g of pure $_{94}Pu^{239}$ undergo fission ? The average energy released per fission is 180 MeV.

CBSE 2024

28. (a) Differentiate between nuclear fission and fusion. **3**

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CBSE 2024

(a) In nuclear fission , a heavy nucleus splits into two or more lighter nuclei and energy is released.	½
In nuclear fusion, lighter nuclei combine together a form a heavy nucleus and larger amount of energy is released.	½
(b) Number of atoms in 1 g of $_{94}Pu^{239}$	
$= \frac{6.023\times10^{23}}{239}$	1
$= 2.5\times10^{21}$	
Energy released in fission of 1 g of $_{94}Pu^{239}$,	
$E = 180MeV\times2.5\times10^{21}$	
$E = 4.5\times10^{23}$ MeV	1

27. (a) Define atomic mass unit (u).

(b) Calculate the energy required to separate a deuteron into its constituent parts (a proton and a neutron). Given : *3*

$m(D) = 2{\cdot}014102$ u

$m_H = 1{\cdot}007825$ u

$m_n = 1{\cdot}008665$ u

CBSE 2024

27. (a) Define atomic mass unit (u).

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$m(D) = 2{\cdot}014102$ u

$m_H = 1{\cdot}007825$ u

$m_n = 1{\cdot}008665$ u

CBSE 2024

(a) Atomic mass unit (u) is defined as $1/12^{th}$ of the mass of the carbon (^{12}C) atom.	**1**
(b) $m\left({}_1H^2\right) \rightarrow m\left({}_1H^1\right) + m\left({}_0n^1\right)$	½
$Q = (m_R - m_P) \times 931.5\,MeV$	
$= (2.014102 - 1.007825 - 1.008665) \times 931.5\,MeV$	½
$= -0.002388 \times 931.5\,MeV$	½
$= -2.224\,MeV$	
Hence energy required is 2.224 MeV	½

26. (a) Briefly discuss three characteristics of the forces between nucleons.

(b) Which out of $^{8}_{4}X$ and $^{5}_{3}Y$ nuclei is more stable and why ? CBSE 2024 *3*

26. (a) Briefly discuss three characteristics of the forces between nucleons.

(b) Which out of $^{8}_{4}X$ and $^{5}_{3}Y$ nuclei is more stable and why ? CBSE 2024 *3*

(a) Characteristics of nuclear forces :- 1. Saturated in nature 2. Attractive for distances larger than r_0 and repulsive for distance less than r_o 3. Do not depend on nature of electric charge i.e. same for n-n, n-p and p-p pairs. 4. Much stronger than gravitational forces. **(Any three)**	**1½**
(b) $^{8}_{4}X$ is more stable	**½**
The ratio of number of neutrons to the number of protons is more in $^{8}_{4}X$ than $^{5}_{3}Y$	**1**

20. Calculate the energy released/absorbed in the following nuclear reaction : *2*

$$^{12}_{6}C + {}^{12}_{6}C \longrightarrow {}^{20}_{10}Ne + {}^{4}_{2}He$$

Given : $m\left({}^{12}_{6}C\right) = 12{\cdot}000000\ u$

$m\left({}^{20}_{10}Ne\right) = 19{\cdot}992439\ u$

$m\left({}^{4}_{2}He\right) = 4{\cdot}002603\ u$

$1\ u = 931\ MeV/c^2$

CBSE 2024

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CBSE 2024

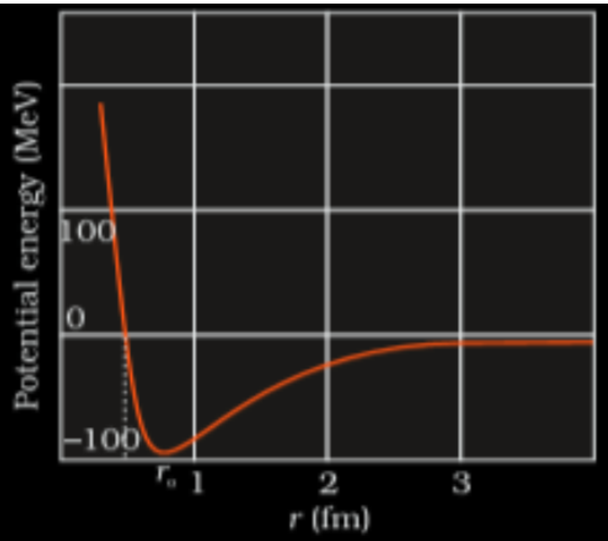
Energy = mass defect x 931 Mev	½
Mass defect $= \Delta m = (2 \times 12.000000 - 19.992439 - 4.002603)$	
$= 0.004958u$	½
Energy released = 0.004958 x 931 MeV	½
= 4.62 MeV	½

27. (a) Plot a graph depicting potential energy of a pair of nucleons in a nucleus as a function of their separation.

(b) Identify the regions where the nuclear force is (i) attractive, and (ii) repulsive. Justify your answer. CBSE 2024 *3*

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(b) Identify the regions where the nuclear force is (i) attractive, and (ii) repulsive. Justify your answer. CBSE 2024 *3*

(a)  (**Give full credit for graph without values**)	**1**
(b) $F = -\frac{dU}{dx}$	
i) For distance larger than r_o, force is attractive	½
Since slope of the curve is positive	½
ii) For distance less than r_o, force is repulsive	½
Since slope of the curve is negative	½

20. The carbon isotope $^{12}_{6}C$ has a nuclear mass of 12·000000 u. Calculate the binding energy of its nucleus.

CBSE 2024

Given $m_p = 1{\cdot}007825$ u; $m_n = 1{\cdot}008665$ u. *2*

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CBSE 2024

Given $m_p = 1{\cdot}007825$ u; $m_n = 1{\cdot}008665$ u. *2*

Binding Energy $= (Zm_p + (A-Z)m_n - M_N) \times 931.5$ MeV	½
B. E.$= (6 \times 1.007825 + 6 \times 1.008665 - 12.000000)$ x 931.5 MeV	½
$= (0.09894)$ x 931.5 MeV	½
B. E. $= 92.16$ MeV	½

26. (a) State any two properties of a nucleus.

(b) Why is the density of a nucleus much more than that of an atom ?

(c) Show that the density of the nuclear matter is the same for all nuclei. CBSE 2024 *3*

26. (a) State any two properties of a nucleus.

(b) Why is the density of a nucleus much more than that of an atom ?

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CBSE 2024 *3*

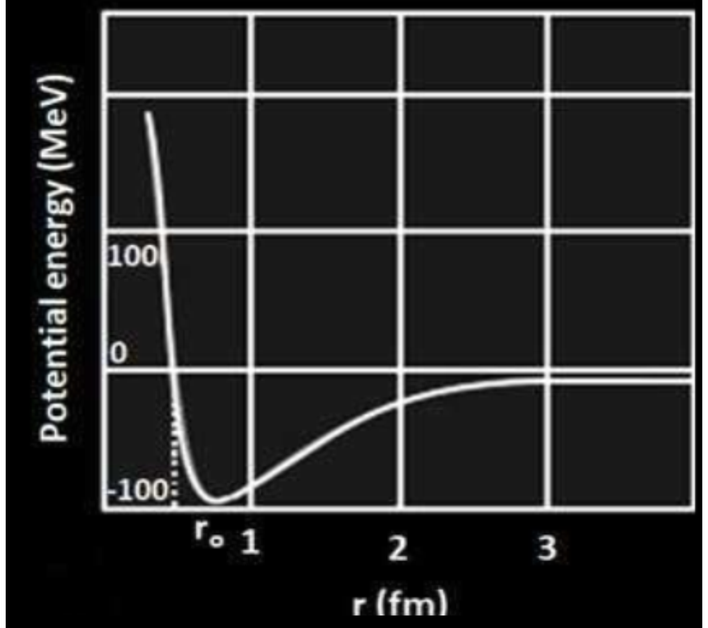
a) (Any TWO) (i) The nucleus is positively charged (ii) The nucleus consists of protons and neutrons (iii) The nuclear density is independent of mass number (iv) The radius of the nucleus, $R = R_o A^{1/3}$	½ + ½
b) Atoms have large amount of empty spaces. Mass is concentrated in nucleus.	1
c) Density = Mass / Volume $= \frac{m\,A}{\frac{4}{3}\pi R^3} = \frac{m\,A}{\frac{4}{3}\pi R_o^3\,A}$ $= \frac{m}{\frac{4}{3}\pi R_o^3}$	
So, density is independent of mass number.	1

28. (a) Write two characteristic properties of nuclear force.

(b) Draw a plot of potential energy of a pair of nucleons as a function of their separation. Write two important conclusions that can be drawn from the plot. CBSE 2024 *3*

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(b) Draw a plot of potential energy of a pair of nucleons as a function of their separation. Write two important conclusions that can be drawn from the plot. CBSE 2024 *3*



(a) Properties of nuclear forces (Any two) :

(i) The nuclear force is much stronger than the Coulomb force acting between charges or the gravitational forces between their masses.

(ii) The nuclear force between two nucleons falls rapidly to zero as their distance becomes more than a few femtometres. — 1

(iii) The nuclear force between neutron- neutron, proton- neutron and proton-proton is approximately same.

(iv) The nuclear force is charge independent. — ½ + ½

Conclusions:-

(i) The potential energy is minimum at a distance r_o.

(ii) The force between the nucleons is attractive for distances larger than r_o and repulsive if they are separated by distance less than r_o. — ½ + ½

20. Differentiate between mass defect and binding energy per nucleon. The binding energy per nucleon is nearly constant in the range $30 < A < 170$. Why ?

CBSE 2024 *2*

20. Differentiate between mass defect and binding energy per nucleon. The binding energy per nucleon is nearly constant in the range $30 < A < 170$. Why ? CBSE 2024 *2*

Mass defect- The difference in mass of a nucleus and total mass of its constituents.	½
Binding energy per nucleon- The average energy per nucleon needed to separate individual nucleons of a nucleus.	½
The constancy of binding energy in the range $30<A<170$ is a consequence of the fact that nuclear forces are short-ranged.	1

23. A nucleus with A = 280 splits into two nuclei X_1 and X_2 whose mass numbers are in the ratio 27 : 8. Find the ratio of their radii, $\left(\frac{r_1}{r_2}\right)$

CBSE 2023 C *2*

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CBSE 2023 C *2*

$R = R_o A^{\frac{1}{3}}$	½
$\frac{R_1}{R_2} = \left(\frac{A_1}{A_2}\right)^{\frac{1}{3}}$	½
$= \left(\frac{27}{8}\right)^{\frac{1}{3}}$	
$\frac{R_1}{R_2} = \frac{3}{2}$	1

29. (a) Obtain the relationship between atomic mass unit (u) and electron volt (eV).

(b) The mass of a ball is 0·5 kg. It is totally converted into energy. Calculate the energy output in eV. CBSE 2023 C *3*

29. (a) Obtain the relationship between atomic mass unit (u) and electron volt (eV).

(b) The mass of a ball is 0·5 kg. It is totally converted into energy. Calculate the energy output in eV. CBSE 2023 C *3*

(a) $E = \Delta m c^2$ ½

$\Delta m = 1.66 \times 10^{-27}\,\text{kg}$

$E = \dfrac{1.66 \times 10^{-27} \times (3 \times 10^8)^2}{1.6 \times 10^{-19}}\,\text{eV}$ ½

$E = 931.5 \times 10^6\,\text{eV}$ ½

(b)

$E = mc^2$ ½

$m = 0.5\,\text{kg}$

$E = (0.5) \times (3 \times 10^8)^2\,\text{J}$

$E = 4.5 \times 10^{16}\,\text{J}$

$E = \dfrac{4.5 \times 10^{16}}{1.6 \times 10^{-19}}\,\text{eV}$ ½

$E = 2.8 \times 10^{23}\,\text{eV}$ ½

8. The binding energy per nucleon is lower : CBSE 2023 C

(a) for light nuclei only

(b) for heavy nuclei only

(c) for nuclei of middle mass numbers only

(d) both for the light nuclei and the heavy nuclei

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11. Two nuclei have their mass numbers in the ratio of 1 : 27. What is the ratio of their nuclear densities ?

(a) 1 : 27 (b) 1 : 1

(c) 1 : 9 (d) 1 : 3

CBSE 2023 C

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(a) 1 : 27 (b) 1 : 1

(c) 1 : 9 (d) 1 : 3

CBSE 2023 C

(b) 1:1

8. The binding energy per nucleon of ${}_8O^{16}$ is 7·97 MeV and that of ${}_8O^{17}$ is 7·75 MeV. The energy (in MeV) required to remove a neutron from ${}_8O^{17}$ is :

(a) 0·42 MeV (b) 7·86 MeV

(c) 4·23 MeV (d) 3·64 MeV

CBSE 2023 C

8. The binding energy per nucleon of $_8O^{16}$ is 7·97 MeV and that of $_8O^{17}$ is 7·75 MeV. The energy (in MeV) required to remove a neutron from $_8O^{17}$ is :

(a) 0·42 MeV (b) 7·86 MeV

(c) 4·23 MeV (d) 3·64 MeV

CBSE 2023 C

(c) 4.23 MeV

8. In the nuclear reaction $_{7}N^{14} + {}_{2}He^{4} \longrightarrow X + {}_{1}H^{1}$, X represents :

(a) $_{7}O^{16}$

(b) $_{8}N^{17}$

(c) $_{8}O^{17}$

(d) $_{7}N^{16}$

CBSE 2023 C

8. In the nuclear reaction $_{7}N^{14} + {}_{2}He^{4} \longrightarrow X + {}_{1}H^{1}$, X represents :

(a) $_{7}O^{16}$ (b) $_{8}N^{17}$

(c) $_{8}O^{17}$ (d) $_{7}N^{16}$

CBSE 2023 C

$(c)\ {}_{8}O^{17}$

CBSE 2023

13. The number of neutrons in $^{237}_{91}Pa$ is :

(a) 91 (b) 146 (c) 237 (d) 164

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CBSE 2023

(a) 91 (b) 146 (c) 237 (d) 164

(b) 146

30. (a) Explain two main features of the plot of the binding energy per nucleon versus the mass number of the nuclei. Mention two conclusions that can be drawn from these features. CBSE 2023 *3*

30. (a) Explain two main features of the plot of the binding energy per nucleon versus the mass number of the nuclei. Mention two conclusions that can be drawn from these features. CBSE 2023 *3*

(i) Binding energy per nucleon ΔE_{bn} is $\approx 8MeV$ for nuclei of middle mass number. It is independent of size of nuclei	1
(ii) ΔE_{bn} is lower for very light nuclei $(A \leq 10)$ and very heavy nuclei $(A \geq 170)$	1
First feature tells us about the strength of nuclear forces.	½
Second feature provide the clues about the stability of the nuclei.	½
Any other features of the graph	

3. Which of the following statements is ***not*** true for nuclear forces ?

(a) They are stronger than Coulomb forces.

(b) They have about the same magnitude for different pairs of nucleons.

(c) They are always attractive.

(d) They saturate as the separation between two nucleons increases.

CBSE 2023

3. Which of the following statements is ***not*** true for nuclear forces ?

(a) They are stronger than Coulomb forces.

(b) They have about the same magnitude for different pairs of nucleons.

(c) They are always attractive.

(d) They saturate as the separation between two nucleons increases.

CBSE 2023

(c) They are always attractive

15. The difference in mass of ^{7}X nucleus and total mass of its constituent nucleons is 21·00 u. The binding energy per nucleon for this nucleus is equal to the energy equivalent of : CBSE 2023

(a) 3 u (b) 3·5 u (c) 7 u (d) 21 u

15. The difference in mass of ^{7}X nucleus and total mass of its constituent nucleons is 21·00 u. The binding energy per nucleon for this nucleus is equal to the energy equivalent of : CBSE 2023

(a) 3 u (b) 3·5 u (c) 7 u (d) 21 u

(a) 3u

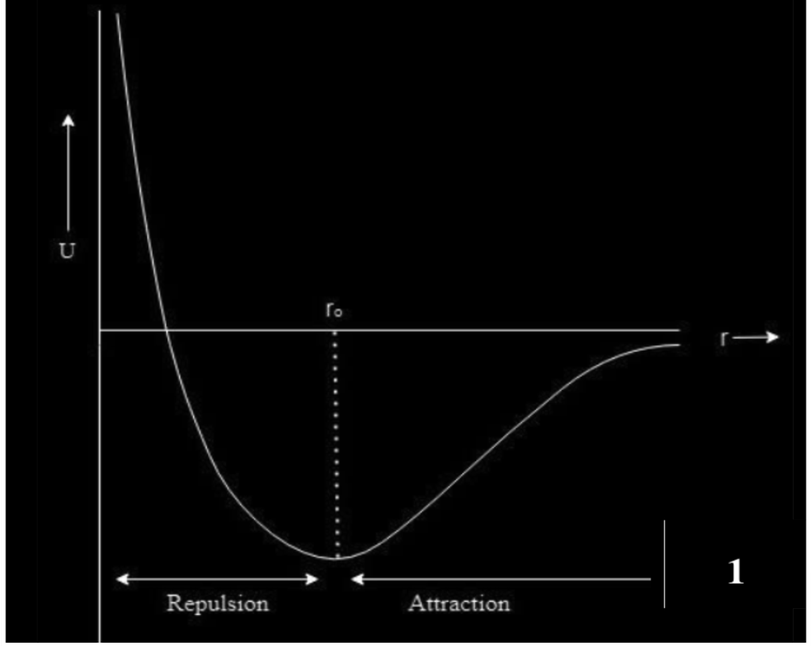
20. Briefly describe the multi-step process involved in the generation of energy in the Sun. CBSE 2023 *2*

The fusion reaction in the sun is a multi- step process in which the hydrogen is burned into helium.	
$H_1^1 + H_1^1 \rightarrow H\,{}_1^2 + e^+ + \upsilon + Q_1$	½
$e^+ + e^- \rightarrow \gamma + \gamma + Q_2$	½
$H_1^2 + H_1^1 \rightarrow He\,{}_2^3 + \gamma + Q_3$	½
$He_2^3 + He_2^3 \rightarrow He\,{}_2^4 + H_1^1 + H_1^1 + Q_4$	½

26. Draw a plot showing the variation of potential energy of two nucleons as a function of distance between them. Identify the regions in which the force between the nucleons is (i) attractive, and (ii) repulsive. Justify your answers. CBSE 2023 *3*

26. Draw a plot showing the variation of potential energy of two nucleons as a function of distance between them. Identify the regions in which the force between the nucleons is (i) attractive, and (ii) repulsive. Justify your answers.

CBSE 2023 *3*



1

Identification:

(i) For a separation greater than r_0, the force is **attractive**. ½

(ii) for a separation less than r_0, the force is **repulsive.** ½

Justification:

The potential energy is a minimum at a distance r_0 . This means that the force is attractive for distance larger than r_0 and repulsive if they are separated by distances less than r_0. ½ ½

Alternatively: $F = -\frac{dU}{dr}$.

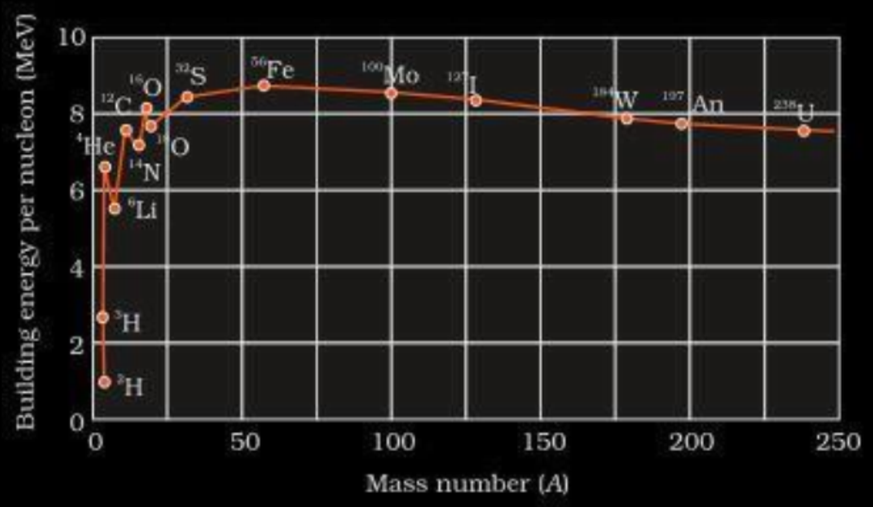
For $r < r_0$,slope is negative and therefore F is positive, i.e. repulsive.

For $r \geq r_0$,slope is positive and therefore F is negative, i.e. attractive.

24. Draw the graph showing the variation of binding energy per nucleon with mass number A of nuclei ($2 < A < 170$). Use this graph to explain the release of energy in nuclear fission. CBSE 2023 *2*

24. Draw the graph showing the variation of binding energy per nucleon with mass number A of nuclei ($2 < A < 170$). Use this graph to explain the release of energy in nuclear fission. CBSE 2023 *2*

Graph :



1

Note: Full marks to be awarded even if values are not marked.

Explanation: A very heavy nucleus has lower binding energy per nucleon compared to that of lighter nuclei. Thus if a heavier nucleus breaks into two nuclei, nucleons get more tightly bound. This implies, energy would be released in the process. 1

Alternatively
In nuclear fission, a heavy nucleus breaks B.E/nucleon increases. So energy is released. 1

16. *Assertion (A):* The nucleus ${}^{7}_{3}\mathrm{X}$ is more stable than the nucleus ${}^{4}_{3}\mathrm{Y}$.

Reason (R): ${}^{7}_{3}\mathrm{X}$ contains more number of protons.

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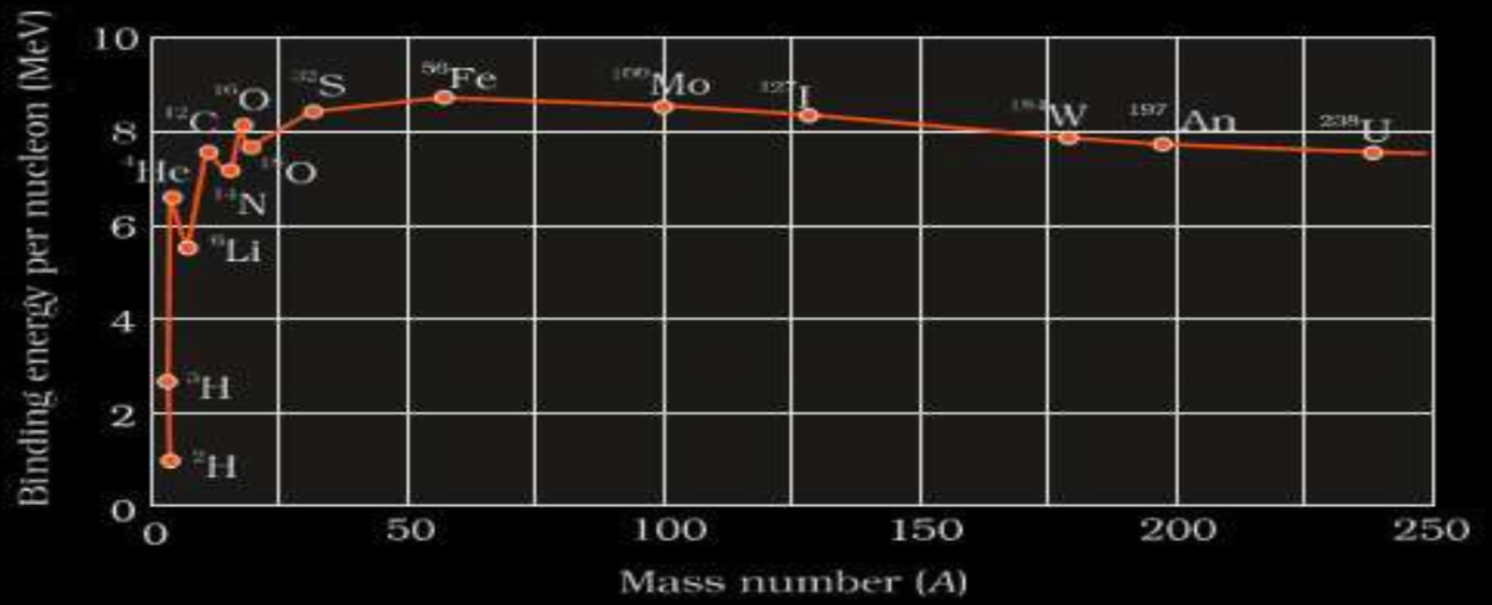
16. *Assertion (A):* The nucleus ${}^{7}_{3}\mathrm{X}$ is more stable than the nucleus ${}^{4}_{3}\mathrm{Y}$.

Reason (R): ${}^{7}_{3}\mathrm{X}$ contains more number of protons. CBSE 2023

(c) Assertion (A) is true but Reason (R) is false

20. (a) Draw a graph showing the variation of binding energy per nucleon as a function of mass number A. The binding energy per nucleon for heavy nuclei (A > 170) decreases with the increase in mass number. Explain. CBSE 2023 *2*

20. (a) Draw a graph showing the variation of binding energy per nucleon as a function of mass number A. The binding energy per nucleon for heavy nuclei (A > 170) decreases with the increase in mass number. Explain. CBSE 2023 *2*



Explanation:-Nuclear forces are short range & show saturation, while the electrostatic force are neither short range nor show any saturation. Hence for heavier nuclei (A>170) the electrostatic force of repulsion becomes predominant, decreasing the binding energy per nucleon. 1

Alternatively:-As the size of the nucleus increases, the nucleus becomes unstable. 1

27. (a) (i) Prove that the nuclear density is same for all nuclei.

(ii) Draw a plot of potential energy of a pair of nucleons as a function of their separation. Draw two inferences from this plot.

CBSE 2023 *3*

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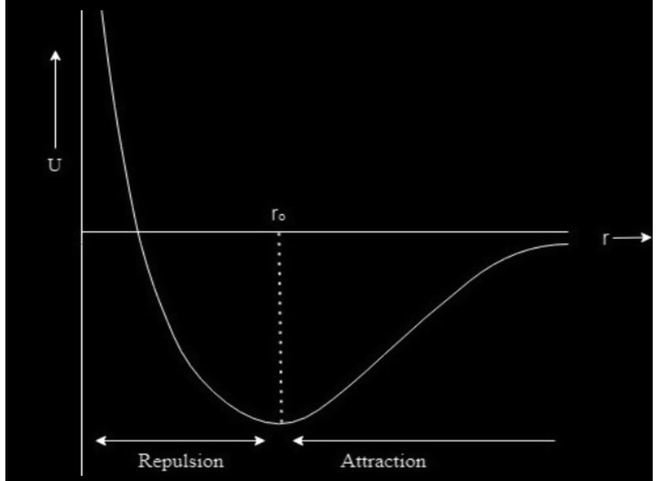
(ii) Draw a plot of potential energy of a pair of nucleons as a function of their separation. Draw two inferences from this plot.

CBSE 2023 *3*

(i) $\rho = \dfrac{mass}{volume}$

$= \dfrac{\text{mas number} \times \text{mass of nucleon}}{volume\ of\ nucleus}$ ½

$\rho = \dfrac{A \times \text{m}}{\frac{4}{3}\pi\left(R_0 A^{1/3}\right)^3} = \dfrac{3m}{4\pi R_o^3}$ ½



Hence, density is independent of mass number.

Inferences

-The force is attractive for distances larger than r_o. ½

- The force is repulsive for distance less than r_o. ½

Alternatively:-

Any other relevant inference drawn from the graph should be given full credit. 1

(b) Draw a diagram to show the variation of binding energy per nucleon with mass number for different nuclei and mention its two features. Why do lighter nuclei usually undergo nuclear fusion ? *3*

CBSE 2023

(b) Draw a diagram to show the variation of binding energy per nucleon with mass number for different nuclei and mention its two features. Why do lighter nuclei usually undergo nuclear fusion ? *3*

CBSE 2023

Features of diagram **(any two)** 1. Binding energy per nucleon is practically independent of atomic number for nuclei of middle mass number (30 <A < 170) 2. The curve has maximum of about 8.75 MeV for A= 56 and has a value of 7.6 MeV for A= 238 3. Binding energy per nucleon is lower for both light nuclei (A<30) and heavy nuclei (A>170)	½ + ½
Two lighter nuclei fuse together to form heavier nuclei as the binding energy per nucleon of fused heavier nuclei is more than the binding energy per nucleon of the lighter nuclei. Thus the final system is more tightly bound than initial system. **Alternatively** To attain the stability	1

6. The atomic number of an atom represents : CBSE 2023

(a) number of neutrons in nucleus.

(b) total number of protons and electrons in the atom.

(c) number of protons in nucleus.

(d) total number of protons and neutrons in nucleus.

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(c) Number of protons in nucleus.

---

12. The mass density of a nucleus of mass number A is :

(a) proportional to $A^{1/3}$

(b) proportional to $A^{2/3}$

(c) proportional to A^3

(d) independent of A

CBSE 2023

12. The mass density of a nucleus of mass number A is :

(a) proportional to $A^{1/3}$

(b) proportional to $A^{2/3}$

(c) proportional to A^3

(d) independent of A

CBSE 2023

(d) Independent of A

6. The radius of $^{27}_{13}X$ nucleus is R. The radius of $^{125}_{53}Y$ nucleus will be **1**

(a) $\frac{5}{3}$ R

(b) $\left(\frac{13}{53}\right)^{1/3}$ R

(c) $\left(\frac{5}{3}R\right)^{1/3}$

(d) $\left(\frac{13}{53}R\right)^{1/3}$

CBSE 2023

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(a) $\frac{5}{3}R$

(b) $\left(\frac{13}{53}\right)^{1/3}R$

(c) $\left(\frac{5}{3}R\right)^{1/3}$

(d) $\left(\frac{13}{53}R\right)^{1/3}$

CBSE 2023

(a) $\left(\frac{5}{3}R\right)$

10. The curve of binding energy per nucleon as a function of atomic mass number has a sharp peak for helium nucleus. This implies that helium nucleus is **1**

(a) radioactive
(b) unstable
(c) easily fissionable
(d) more stable nucleus than its neighbours

CBSE 2023

10. The curve of binding energy per nucleon as a function of atomic mass number has a sharp peak for helium nucleus. This implies that helium nucleus is **1**

(a) radioactive
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(d) more stable nucleus than its neighbours

CBSE 2023

(d) More stable nucleus than its neighbours.

(b) Define the term, mass defect. How is it related to stability of the nucleus ?

CBSE 2023

(b) Define the term, mass defect. How is it related to stability of the nucleus ?

CBSE 2023

Mass defect is the difference between the actual mass of the nucleus and the sum of the masses of its nucleons.	1
Greater the mass defect, greater will be the binding energy and the nucleus will be more stable	1
Alternatively	
Give full credit (1 mark) if a candidate writes, mass defect $\propto$ stability of the nucleus.	

(a) (i) Distinguish between nuclear fission and fusion giving an example of each.

(ii) Explain the release of energy in nuclear fission and fusion on the basis of binding energy per nucleon curve. CBSE 2023 **3**

(a) (i) Distinguish between nuclear fission and fusion giving an example of each.

(ii) Explain the release of energy in nuclear fission and fusion on the basis of binding energy per nucleon curve. CBSE 2023 **3**

Nuclear fission – It is a process in which a heavy nucleus when excited (say on bombarding by a slow moving neutron) splits into two lighter nuclei of nearly comparable masses with a release of large amount of energy.	½
Example of nuclear fission $_0^1n + {}_{92}^{235}U \rightarrow {}_{92}^{236}U \rightarrow {}_{56}^{144}Ba + {}_{36}^{89}Kr + 3\,{}_0^1n + Q$	½
Nuclear Fusion - It is a process in which two lighter nuclei fuse (at extremely high temperature) to form a heavy nucleus and large amount of energy is released.	½
Examples of nuclear fusion (i) $_1^1H + {}_1^1H \rightarrow {}_1^2H + e^+ + \nu + Q_1$ (ii) $_1^2H + {}_1^2H \rightarrow {}_2^3H + n + Q_2$ (iii) $_1^2H + {}_1^2H \rightarrow {}_1^3H + {}_1^1H + Q_3$ (any other possible reaction equation)	½
(ii) The binding energy per nucleon of the products in the nuclear reactions (nuclear fission and nuclear fusion) is greater than that of the reactants .	1

(b) (i) How is the size of a nucleus found experimentally ? Write the relation between the radius and mass number of a nucleus.

(ii) Prove that the density of a nucleus is independent of its mass number.

CBSE 2023

(b) (i) How is the size of a nucleus found experimentally ? Write the relation between the radius and mass number of a nucleus.

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CBSE 2023

(i) Size of nucleus of an atom is determined by scattering experiments in which fast electrons are used to bombard targets.	½
Relation between radius and mass number of nucleus. $R = R_0 A^{1/3}$	
(ii) Density of nucleus	1
$\rho = \frac{mass}{volume}$	½
$\rho = \frac{m \times A}{\frac{4}{3}\pi R^3}$	
$\rho = \frac{mA}{\frac{4}{3}\pi\left(R_0 A^{1/3}\right)^3}$	½
$\rho = \frac{3m}{4\pi R_0^{\,3}}$	
Hence, density of nucleus is independent of mass number (A).	½

8. The ratio of the nuclear densities of two nuclei having mass numbers 64 and 125 is **1**

(A) $\frac{64}{125}$

(B) $\frac{4}{5}$

(C) $\frac{5}{4}$

(D) 1

CBSE 2023

8. The ratio of the nuclear densities of two nuclei having mass numbers 64 and 125 is **1**

(A) $\frac{64}{125}$

(B) $\frac{4}{5}$

(C) $\frac{5}{4}$

(D) 1

CBSE 2023

(D) 1

30. (a) Calculate the binding energy of an alpha particle in MeV. Given **3**

mass of a proton = 1.007825 u

mass of a neutron = 1.008665 u

mass of He nucleus = 4.002800 u

$1u = 931\ MeV/c^2$

CBSE 2023

30. (a) Calculate the binding energy of an alpha particle in MeV. Given **3**

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mass of a neutron = 1.008665 u
mass of He nucleus = 4.002800 u
$1u = 931\ MeV/c^2$

CBSE 2023

Number of Protons = 2 Number of neutrons = 2 $B.E. = \left[\left(ZM_P + (A-Z)m_n\right) - m_N\right]C^2$	**1**
$\Delta m = [(2 \times 1.007825 + 2 \times 1.008665) - (4.002800)]\ u$	**1**
B. E. $= 0.03018 \times 931$ MeV	
B E. = 28 MeV	**1**

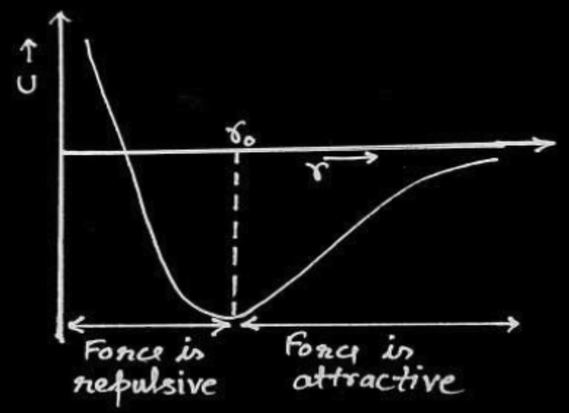
(b) A heavy nucleus P of mass number 240 and binding energy 7.6 MeV per nucleon splits into two nuclei Q and R of mass number 110 and 130 and binding energy per nucleon 8.5 MeV and 8.4 MeV respectively. Calculate the energy released in the fission. CBSE 2023 **3**

(b) A heavy nucleus P of mass number 240 and binding energy 7.6 MeV per nucleon splits into two nuclei Q and R of mass number 110 and 130 and binding energy per nucleon 8.5 MeV and 8.4 MeV respectively. Calculate the energy released in the fission. CBSE 2023 **3**

Binding energy of nucleus P $= 7.6 \times 240 = 1824$ MeV	½
Binding energy of nucleus Q $= 8.5 \times 110 = 935$ MeV	½
Binding energy of nucleus R $= 8.4 \times 130 = 1092$ MeV	½
Binding energy of (Q + R) = 935 + 1092 = 2027 MeV	½
Energy released = Binding Energy of (Q + R) - Binding energy of P	½
Energy released = (2027- 1824) MeV	½
= 203 MeV	

19. Draw a graph showing the variation of potential energy of a pair of nucleons as a function of their separation. Indicate the region in which the nuclear force is (a) attractive and (b) repulsive. CBSE 2023 **2**

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1

Repulsive nuclear force for $r < r_0$ **½**

Attractive force for $r > r_0$ **½**

(Note- If student draws a graph without indicating the region of attractive and repulsive force, give 1 mark.)

2. (a) Briefly explain how energy is released due to proton-proton cycle in the interior of the Sun. CBSE 2022 C *2*

2. (a) Briefly explain how energy is released due to proton-proton cycle in the interior of the Sun. CBSE 2022 C *2*

Nuclear fusion is the source of energy generation in the proton- proton cycle taking place in the sun.	1
In the core of the sun, due to extremely high temperature, hydrogen nuclei fuse together and form helium nuclei releasing thermonuclear energy.	1
Alternatively Equations depicting fusion of four hydrogen nuclei i.e $4_1^1H + 2e^- \rightarrow {}_2^4He + 2\nu + 6\gamma + 26.7\,MeV$ or $(4_1^1H + 4e^-) \rightarrow ({}_2^4He + 2e^-) + 2\nu + 6\gamma + 26.7\,MeV$	

6. (a) Prove that different nuclei have the same density.

(b) Find the radius of a nucleus with mass number 64. (R_o = 1·2 fm)

CBSE 2022 C *3*

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CBSE 2022 C

3

(a) Mass of the nucleus= m A Nuclear density $= \frac{\text{Mass}}{\text{Volume}}$	½
$= \frac{m\,A}{\frac{4}{3}\pi R^3}$	½
But $R = R_0 A^{1/3}$	½
Nuclear density $= \frac{m}{\frac{4}{3}\pi R_0^3}$ which is constant	½
(b) $R = R_0 A^{1/3} = 1.2\,(64)^{1/3}$	½
$R = 1.2 \times 4 = 4.8$ fm	½

7. (a) Determine from the given data, whether the following reaction is exothermic or endothermic : CBSE 2022 C *2*

$$^{226}_{88}\text{Ra} \longrightarrow {}^{222}_{86}\text{Rn} + {}^{4}_{2}\text{He}$$

Atomic masses : $m\left(^{226}_{88}\text{Ra}\right) = 226{\cdot}02540\ \text{u}$

$m\left(^{222}_{86}\text{Rn}\right) = 222{\cdot}01750\ \text{u}$

$m\left(^{4}_{2}\text{He}\right) = 4{\cdot}002603\ \text{u}$

(b) What is the ratio of the nuclear density of $^{4}_{2}\text{He}$ and $^{14}_{7}\text{N}$? *1*

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(b) What is the ratio of the nuclear density of ${}^{4}_{2}\text{He}$ and ${}^{14}_{7}\text{N}$? *1*

(a) mass of reactant = 226.02540 u mass of the products = (222.01750+4.002603)u = (226.020103)u	1
∴Mass of reactants > Mass of products Hence the reaction is exothermic	½ ½
(b) 1:1	1

9. (a) Determine from the given data, whether the following reaction is exothermic or endothermic : *2*

CBSE 2022 C

$$_{1}^{1}H + {}_{1}^{3}H \longrightarrow {}_{1}^{2}H + {}_{1}^{2}H$$

Atomic masses : $m\left({}_{1}^{2}H\right) = 2{\cdot}014102\ u$

$m\left({}_{1}^{3}H\right) = 3{\cdot}016049\ u$

$m\left({}_{1}^{1}H\right) = 1{\cdot}007825\ u$

(b) What is the ratio of the nuclear density of the gold isotope $_{79}^{197}Au$ and the silver isotope $_{47}^{107}Ag$. *1*

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CBSE 2022 C

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a) Mass of reactants = (1.007825+3.016049) u = 4.023874 u	½
Mass of products = (2x 2.014102) u = 4.028204 u	½
Mass of reactants < Mass of products	½
Hence, reaction is Endothermic	½
b) 1:1	1

10. (a) Determine from the given data, whether the following reaction is exothermic or endothermic : CBSE 2022 C *2*

$$^{12}_{6}C + {}^{12}_{6}C \longrightarrow {}^{20}_{10}Ne + {}^{4}_{2}He$$

Atomic masses : $m\left({}^{4}_{2}He\right) = 4{\cdot}002603\ u$

$m\left({}^{12}_{6}C\right) = 12{\cdot}000000\ u$

$m\left({}^{20}_{10}Ne\right) = 19{\cdot}992439\ u$

(b) What is the ratio of the nuclear density of the phosphorus ${}^{30}_{15}P$ and nitrogen ${}^{14}_{7}N$? *1*

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$m\left({}^{20}_{10}Ne\right) = 19{\cdot}992439\ u$

(b) What is the ratio of the nuclear density of the phosphorus $^{30}_{15}P$ and nitrogen $^{14}_{7}N$? *1*

(a) Mass of the products = 19.992439 + 4.002603 = 23.995042 u	½
Mass of the reactant = (2 × 12.000000) u = 24.00000 u	½
As the mass of the reactants is more than the mass of the products, the reaction is exothermic.	½ +½
(b) 1:1	1

8. In a fission event of $^{238}_{92}U$ by fast moving neutrons, no neutrons are emitted and final products, after the beta decay of the primary fragments, are $^{140}_{58}Ce$ and $^{99}_{44}Ru$. Calculate Q for this process. Neglect the masses of electrons/ positrons emitted during the intermediate steps. 3

Given : $m\left(^{238}_{92}U\right) = 238.05079u$; $m\left(^{140}_{58}Ce\right) = 139.90543u$

$m\left(^{99}_{44}Ru\right) = 98.90594u$; $m\left(^{1}_{0}n\right) = 1.008665u$

CBSE 2022

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CBSE 2022

$\Delta m =$ total mass of the reactants – total mass of the products	½
$= [m(^{238}_{92}U) + m_n - m(^{140}_{58}Ce) - m(^{99}_{44}Ru)]$	½
$= [238 \cdot 05079 + 1 \cdot 008665 - 139 \cdot 90543 - 98 \cdot 90594]u$	½
$= [239 \cdot 059455 - 238 \cdot 81137]u$	
$= 0 \cdot 248085\, u$	½
Q-value $= 0 \cdot 248085$ x 931.5 MeV	
= 231.09 MeV	1
(**Note:** Award this 1 mark even if Q-value is not calculated)	

1. (a) (i) Distinguish between isotopes and isobars. **2**

(ii) Two nuclei have different mass numbers A_1 and A_2. Are these nuclei necessarily the isotopes of the same element ? Explain.

CBSE 2022

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(ii) Two nuclei have different mass numbers A_1 and A_2. Are these nuclei necessarily the isotopes of the same element ? Explain.

CBSE 2022

(i) Isotopes – These are the atoms having same atomic number (Z) but different atomic mass(A).	½
Isobars – The atoms of different element having same atomic masses.	½
(ii) No, the mass number of a nucleus is the sum of number of proton(Z) and number of neutrons (N) / $A = Z + N$ / Two nuclei with different mass numbers A_1 and A_2 , may have , have different Z .	½ + ½

10. (a) James Chadwick, in 1932, studied the emission of neutral radiations when Beryllium nuclei were bombarded with alpha particles. He concluded that emitted radiations were neutrons and not photons. Explain.

(b) Two nuclei may have the same radius, even though they contain different number of protons and neutrons. Explain. CBSE 2022 **3**

10. (a) James Chadwick, in 1932, studied the emission of neutral radiations when Beryllium nuclei were bombarded with alpha particles. He concluded that emitted radiations were neutrons and not photons. Explain.

(b) Two nuclei may have the same radius, even though they contain different number of protons and neutrons. Explain. CBSE 2022 **3**

a) If the neutral radiation consisted of photons then from the conservation of energy and momentum, the energy of photons would have to be much higher than the energy available from the bombardment of beryllium nuclei with α- particles. He assumed that the neutral radiation consists of a new type of neutral particles called neutron	1 ½
b) Radius R of a nucleus depends on mass number A, $R \alpha A^{\frac{1}{3}}$	½
Where $A = N + Z$	½
In case of Isobars mass number is same but N and Z are not equal.	½

8. Calculate the energy released in MeV in the following reaction : **3**

$$_{1}^{2}H + _{1}^{3}H \longrightarrow _{2}^{4}He + n$$

Given : $m\left(_{1}^{2}H\right) = 2.014102\ u$

$m\left(_{1}^{3}H\right) = 3.016049\ u$

$m\left(_{2}^{4}He\right) = 4.002603\ u$

$m_n = 1.008665\ u$

8. Calculate the energy released in MeV in the following reaction : **3**

CBSE 2022

$$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \longrightarrow {}^{4}_{2}\mathrm{He} + \mathrm{n}$$

Given : $m\left({}^{2}_{1}\mathrm{H}\right) = 2.014102\ \mathrm{u}$

$m\left({}^{3}_{1}\mathrm{H}\right) = 3.016049\ \mathrm{u}$

$m\left({}^{4}_{2}\mathrm{He}\right) = 4.002603\ \mathrm{u}$

$m_n = 1.008665\ \mathrm{u}$

$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + n + Energy$

Mass defect = mass of reactants – mass of products ½

$\Delta m = m({}^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H}) - m({}^{4}_{2}\mathrm{He} + {}^{1}_{0}n)$ ½

Mass defect $= (2\cdot014102 + 3\cdot016049) - (4\cdot002603 + 1\cdot008665)$ ½

$= 5\cdot030151 - 5\cdot011268$ ½

$= 0\cdot018883u$ ½

Energy released = $\Delta m \times 931.5$ MeV ½

$= 0.018883 \times 931.5$ MeV

$= 17\cdot58$ MeV ½

5. (a) Differentiate between nuclear fission and nuclear fusion.

(b) Deuterium undergoes fusion as per the reaction :

$$_{1}^{2}\mathrm{H} + {}_{1}^{2}\mathrm{H} \longrightarrow {}_{2}^{3}\mathrm{He} + {}_{0}^{1}\mathrm{n} + 3{\cdot}27 \text{ MeV}$$

Find the duration for which an electric bulb of 500 W can be kept glowing by the fusion of 100 g of deuterium. CBSE 2022 *3*

5. (a) Differentiate between nuclear fission and nuclear fusion.

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Find the duration for which an electric bulb of 500 W can be kept glowing by the fusion of 100 g of deuterium. CBSE 2022 *3*

a) Nuclear fission – The process of breaking a very heavy nucleus into lighter nuclei, having mass number in the range of middle mass number ($30 < A < 170$). ½

Nuclear Fusion- It is the process of joining of very light nuclei ($A \leq 10$) to form a heavier nucleus. ½

b) No. of atoms in 100 g $= \dfrac{6{\cdot}023\times10^{23}}{2}\times100 = 3{\cdot}0115\times10^{25}$ ½

Energy released/atom $= \dfrac{3{\cdot}27\ \text{MeV}}{2} = 1{\cdot}635\ \text{MeV}$

Total energy released $= 3{\cdot}0115\times10^{25}\times1{\cdot}635\ \ \text{MeV}$

$= 3{\cdot}0115\times1{\cdot}635\times10^{25}\times1{\cdot}6\times10^{-13}$

$= 7{\cdot}878\times10^{12}\ \text{J}$ ½

$t = \dfrac{E}{P}$ ½

$= \dfrac{7{\cdot}878\times10^{12}\ \text{J}}{500\ \text{J/s}} = 1.5756\times10^{10}\ \text{s}$ ½

$= \dfrac{1.5756\times10^{10}}{3{\cdot}15\times10^{7}} \simeq 500y$

4. (a) Depict the variation of the potential energy of a pair of nucleons with the separation between them.

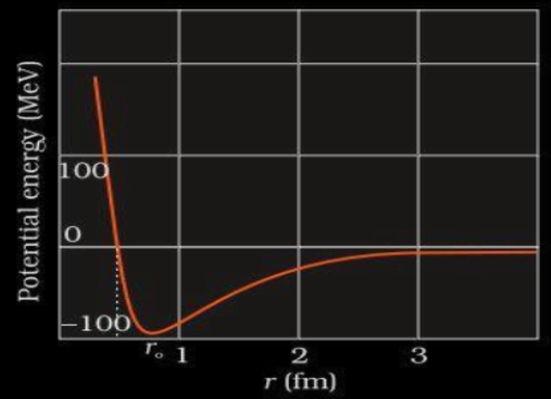
(b) Imagine the fission of a $^{56}_{26}\text{Fe}$ into two equal fragments of $^{28}_{13}\text{Al}$ nucleus. Is the fission energetically possible ? Justify your answer by working out Q value of the process. CBSE 2022 *3*

Given : m $\left(^{56}_{26}\text{Fe}\right)$ = 55·93494 u, m $\left(^{28}_{13}\text{Al}\right)$ = 27·98191 u.

4. (a) Depict the variation of the potential energy of a pair of nucleons with the separation between them.

(b) Imagine the fission of a $^{56}_{26}\text{Fe}$ into two equal fragments of $^{28}_{13}\text{Al}$ nucleus. Is the fission energetically possible ? Justify your answer by working out Q value of the process. CBSE 2022 *3*

Given : m $\left(^{56}_{26}\text{Fe}\right)$ = 55·93494 u, m $\left(^{28}_{13}\text{Al}\right)$ = 27·98191 u.

(a)  	1 ½
(b)Mass Difference = 55.93494 – 2 X 27.98191	1
= - 0.02442 u	
Fission not possible.	½

24. A heavy nucleus P of mass number 240 and binding energy 7.6 MeV per nucleon splits in to two nuclei Q and R of mass numbers 110, 130 and binding energy per nucleon 8.5 MeV and 8.4 MeV, respectively. Calculate the energy released in the fission. CBSE 2020 **2**

24. A heavy nucleus P of mass number 240 and binding energy 7.6 MeV per nucleon splits in to two nuclei Q and R of mass numbers 110, 130 and binding energy per nucleon 8.5 MeV and 8.4 MeV, respectively. Calculate the energy released in the fission. CBSE 2020 **2**

B.E. of heavy nucleus P	$= 240 \times 7.6$ MeV$=1824$ MeV	½
B.E. of nucleus Q	$= 110 \times 8.5$ MeV$=935$ MeV	½
B.E. of nucleus R	$= 130 \times 8.4$ MeV$=1092$ MeV	½
Energy released		
	$= [(935 + 1092) - 1824]$ MeV	
	$= [2027 - 1824]$ MeV	
	$= 203$ MeV	½

15. Name the particle emitted spontaneously in the following nuclear reaction :

$$^{32}_{15}\mathrm{P} \longrightarrow {}^{32}_{16}\mathrm{S} + \overline{v} + _____$$

CBSE 2020 1

15. Name the particle emitted spontaneously in the following nuclear reaction :

$${}^{32}_{15}\mathrm{P} \longrightarrow {}^{32}_{16}\mathrm{S} + \overline{v} + _____$$

CBSE 2020 **1**

$\underline{\beta^- / e^{-1} / electron}$

31. (a) The density of the nuclear matter is tremendously larger than the physical density of the material. Explain.

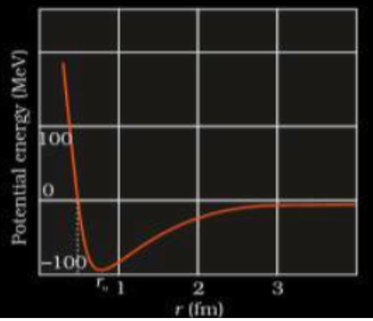
(b) The nuclear forces are not coulomb forces between nucleons. Explain.

(c) Draw a plot of the potential energy between a pair of nucleons as a function of distance between them inside a nucleus. CBSE 2020 *3*

31. (a) The density of the nuclear matter is tremendously larger than the physical density of the material. Explain.

(b) The nuclear forces are not coulomb forces between nucleons. Explain.

(c) Draw a plot of the potential energy between a pair of nucleons as a function of distance between them inside a nucleus. CBSE 2020 *3*

a) Volume of Nucleus is very small but its mass is almost the total mass of the atom $$Now \quad density = \frac{Mass}{Volume}$$ That is why density of nucleus is very high. **Alternatively**, the matter consisting of atoms, has a very large amount of empty space.	1
b) Nuclear forces are very strong, attractive and independent of charge and are short ranged. Whereas Colombian Force are charge dependent and long range. (Accept any one point of difference)	1
	1

9. When two nuclei ($A \leq 10$) fuse together to form a heavier nucleus, the *1*

(A) binding energy per nucleon increases.

(B) binding energy per nucleon decreases.

(C) binding energy per nucleon does not change.

(D) total binding energy decreases.

CBSE 2020

9. When two nuclei ($A \leq 10$) fuse together to form a heavier nucleus, the *1*

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CBSE 2020

(A) binding energy per nucleon increases

10. In β^- decay, a *1*

(A) neutron converts into a proton emitting antineutrino.

(B) neutron converts into a proton emitting neutrino.

(C) proton converts into a neutron emitting antineutrino.

(D) proton converts into a neutron emitting neutrino.

CBSE 2020

10. In β^- decay, a *1*

(A) neutron converts into a proton emitting antineutrino.

(B) neutron converts into a proton emitting neutrino.

(C) proton converts into a neutron emitting antineutrino.

(D) proton converts into a neutron emitting neutrino.

CBSE 2020

(A) neutron converts into a proton emitting antineutrino.

In β-decay, the parent and daughter nuclei have the same number of _____ .

CBSE 2020

1

In β-decay, the parent and daughter nuclei have the same number of _Nucleons_ .

CBSE 2020

1

18. The nuclear radius of $^{27}_{13}\text{Al}$ is 3·6 fermi. Find the nuclear radius of $^{64}_{29}\text{Cu}$. *1*

CBSE 2020

18. The nuclear radius of $^{27}_{13}\text{Al}$ is 3·6 fermi. Find the nuclear radius of $^{64}_{29}\text{Cu}$. *1*

CBSE 2020

4.8 fermi

33. The nucleus $^{235}_{92}Y$, initially at rest, decays into $^{231}_{90}X$ by emitting an α-particle

CBSE 2020

$$^{235}_{92}Y \longrightarrow {}^{231}_{90}X + {}^{4}_{2}He + \text{energy}.$$

The binding energies per nucleon of the parent nucleus, the daughter nucleus and α-particle are 7·8 MeV, 7·835 MeV and 7·07 MeV, respectively. Assuming the daughter nucleus to be formed in the unexcited state and neglecting its share in the energy of the reaction, find the speed of the emitted α-particle. (Mass of α-particle = $6{\cdot}68 \times 10^{-27}$ kg) *3*

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KE of α particle $E_{k\alpha} = (m_y - m_x - m_\alpha)c^2$	½
$= m_y c^2 - m_x c^2 - m_\alpha c^2$	½
$= (235 \times 7.8 - 231 \times 7.835 - 4 \times 7.07)$ MeV	½
$= 1833 - 1809.885 - 28.28$	½
$= 1833 - 1838.165 = -5.165$ MeV	1
$E_k < 0$ wrong information	
[Award full marks till this step]	

(a) Distinguish between isotopes and isobars, giving one example for each.

(b) Why is the mass of a nucleus always less than the sum of the masses of its constituents ? Write one example to justify your answer.

CBSE 2019

(a) Isotopes have same atomic number & isobars have same mass number	½ + ½
Examples of Isotopes $^{12}_{6}C$, $^{14}_{6}C$ Examples Isobars $^{3}_{2}He$, $^{3}_{1}H$	½ + ½
(b) Mass of a nucleus is less than its constituents because in the bound state. some mass is converted into binding energy which is energy equivalent of mass defect e.g. mass of $^{16}_{8}O$ nucleus is less than the sum of masses of 8 protons and 8 neutrons	½ + ½

(a) Classify the following six nuclides into (i) isotones, (ii) isotopes, and (iii) isobars :

$^{12}_{6}C$, $^{3}_{2}He$, $^{198}_{80}Hg$, $^{3}_{1}H$, $^{197}_{79}Au$, $^{14}_{6}C$

(b) How does the size of a nucleus depend on its mass number ? Hence explain why the density of nuclear matter should be independent of the size of the nucleus.

CBSE 2019

(a) Classify the following six nuclides into (i) isotones, (ii) isotopes, and (iii) isobars :

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CBSE 2019

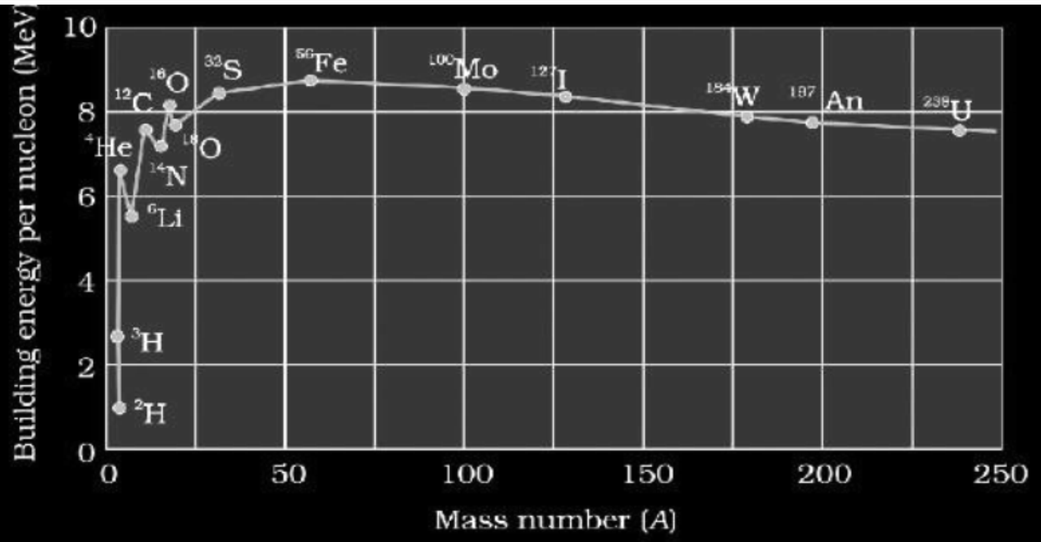
(i) Isotones: $^{198}_{80}Hg$ & $^{197}_{79}Au$	½
(ii)Isotopes: $^{12}_{6}C$, $^{14}_{6}C$	½
(iii) For Isobars: $^{3}_{2}He$, $^{3}_{1}H$	½
The radius of a nucleus having mass number A is $R = r_o A^{1/3}$ r_o is constant.	½
Volume of the nucleus= $\frac{4}{3}\pi R^3 = \frac{4}{3}\pi(r_o A^{1/3})^3$ $= \frac{4}{3}\pi(r_o)^3 A$	½
If 'm' be the average mass of a nucleon then mass of the nucleus= mA	
Nuclear density = $\frac{mass}{Volume} = \frac{mA}{\frac{4}{3}\pi(r_o)^3 A} = \frac{3m}{4\pi r_o^3}$	½
i.e. nuclear density is independent of the size of the nucleus.	

Draw a graph showing the variation of binding energy per nucleon with mass number of different nuclei. Write any two salient features of the curve. How does this curve explain the release of energy both in the processes of nuclear fission and fusion ?

CBSE 2019

Draw a graph showing the variation of binding energy per nucleon with mass number of different nuclei. Write any two salient features of the curve. How does this curve explain the release of energy both in the processes of nuclear fission and fusion ?

CBSE 2019



1

Salient feature of B.E. curve

i) B.E/nucleon is pratically constant i.e. independent of the atomic number for nuclei of middle mass number (30<A<17) ½

ii) Binding energy per nucleon is lower for both light nuclei (A<30) and heavy nuclei (A>170) ½

Very heavy nucleus has lower B.E./nucleon will undergo fission and split into two medium sized nuclei with large B.E./nucleon and release tremendous amount of energy (Fission process) ½

When two very light nuclei, having low binding energy per nucleon combine together and form a medium sized nuclei of higher B.E. per nucleon releases enormous amount of energy (Fusion process) ½

(a) Write two distinguishing features of nuclear forces.

(b) Complete the following nuclear reactions for α and β decay :

(i) $^{238}_{92}\mathrm{U} \longrightarrow ? + {}^{4}_{2}\mathrm{He} + \mathrm{Q}$

(ii) $^{22}_{11}\mathrm{Na} \longrightarrow {}^{22}_{10}\mathrm{Ne} + ? + \nu$

CBSE 2019

(a) Write two distinguishing features of nuclear forces.

(b) Complete the following nuclear reactions for α and β decay :

(i) $^{238}_{92}U \longrightarrow ? + {}^{4}_{2}He + Q$

(ii) $^{22}_{11}Na \longrightarrow {}^{22}_{10}Ne + ? + \nu$

CBSE 2019

i) Any two points given :
(Nuclear forces are charge independent, Non central, spin dependent, strong force, short ranged.)

ii) $^{238}_{92}U \longrightarrow {}^{234}_{90}Th + {}^{4}_{2}He + Q$

(Award this ½ mark even if the student writes the first term as $^{234}_{90}X$)

$^{22}_{11}Na \longrightarrow {}^{22}_{10}Ne + {}^{0}_{+1}e + \nu$

(a) State two distinguishing features of nuclear force.

(b) Draw a plot showing the variation of potential energy of a pair of nucleons as a function of their separation. Mark the regions on the graph where the force is (i) attractive, and (ii) repulsive. CBSE 2019

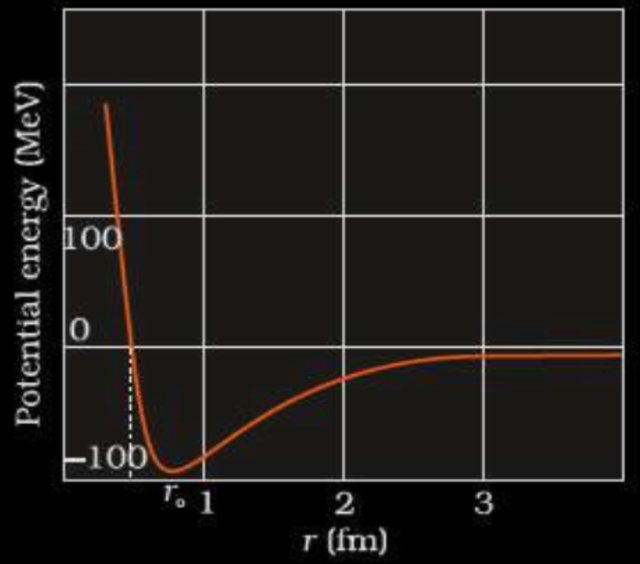
(a) State two distinguishing features of nuclear force.

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CBSE 2019

(1) Short rang force
(2) Strongest force
(3) Attractive in nature
(4) Does not depend on charge (any two) — 1

(b) Figure



1

$r < r_0$ repulsive force — ½

$r > r_0$ attractive force — ½

(a) Draw a plot showing the variation of potential energy of a pair of nucleons as a function of their separation. Mark the regions where the nuclear force is (i) attractive and (ii) repulsive.

(b) In the nuclear reaction

$$n + {}^{235}_{92}U \longrightarrow {}^{a}_{54}Xe + {}^{94}_{b}Sr + 2n$$

determine the values of a and b.

Asha's mother read an article in the newspaper about a disaster that took place at Chernobyl. She could not understand much from the article and asked a few questions from Asha regarding the article. Asha tried to answer her mother's questions based on what she learnt in Class XII Physics.

(a) What was the installation at Chernobyl where the disaster took place ? What, according to you, was the cause of this disaster ?

(b) Explain the process of release of energy in the installation at Chernobyl.

(c) What, according to you, were the values displayed by Asha and her mother ?

CBSE 2017

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CBSE 2017

a)	(i) Nuclear Power Plant:/'Set-up' for releasing Nuclear Energy/Energy Plant (Also accept any other such term)	½
	(ii)Leakage in the cooling unit/ Some defect in the set up.	½
b)	Nuclear Fission/Nuclear Energy Break up (/ Fission) of Uranium nucleus into fragments	1
c)	Asha: Helpful, Considerate, Keen to Learn, Modest	1
	Mother: Curious, Sensitive, Eager to Learn, Has no airs (Any one such value in each case)	1

A nucleus with mass number A = 240 and BE/A = 7.6 MeV breaks into two fragments each of A = 120 with BE/A = 8.5 MeV. Calculate the released energy.

CBSE 2016

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Binding energy of nucleus with mass number 240, E_{bn} = 240 × 7.6 MeV	½
Binding energy of two fragments = 2 × 120 × 8.5 MeV	½
Energy released = 240 (8.5 – 7.6) MeV = 240 × 0.9 = 216 MeV	½ ½

Calculate the energy in fusion reaction :

CBSE 2016

$_{1}^{2}H + _{1}^{2}H \longrightarrow _{2}^{3}He + n$, where BE of $_{1}^{2}H = 2.23$ MeV and of $_{2}^{3}He = 7.73$ MeV.

Calculate the energy in fusion reaction :

CBSE 2016

$^{2}_{1}H + {}^{2}_{1}H \longrightarrow {}^{3}_{2}He + n$, where BE of $^{2}_{1}H = 2.23$ MeV and of $^{3}_{2}He = 7.73$ MeV.

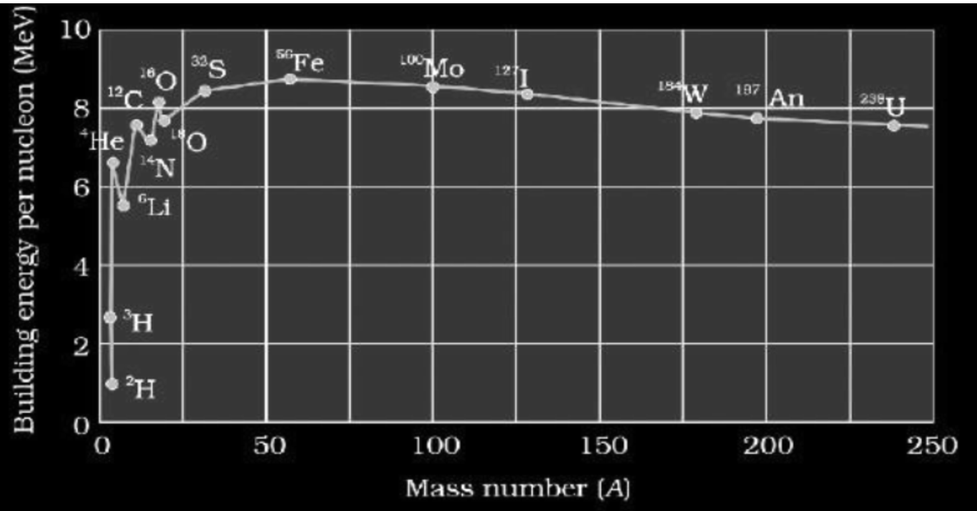
Total Binding energy of Initial System	
i.e. $^{2}_{1}H + {}^{2}_{1}H = (2.23 + 2.23)$ MeV $= 4.46$ MeV	½
Binding energy of Final System i.e. $^{3}_{2}He$ $= 7.73$ MeV	½
Hence energy released = 7.73 MeV- 4.46 MeV $= 3.27$ MeV	1

Plot a graph showing the variation of binding energy per nucleon as a function of mass number. Which property of nuclear force explains the approximate constancy of binding energy in the range $30 < A < 170$? How does one explain the release of energy in both the processes of nuclear fission and fusion from the graph ?

CBSE 2016

Plot a graph showing the variation of binding energy per nucleon as a function of mass number. Which property of nuclear force explains the approximate constancy of binding energy in the range $30 < A < 170$? How does one explain the release of energy in both the processes of nuclear fission and fusion from the graph ?

CBSE 2016



1

Nuclear force is Saturated, or short ranged [any one]	½
The final system is more tightly bound when heavy nucleus undergoes nuclear fission. Hence, there is a release of energy.	½
The final system is more tightly bound when light nuclei undergoes nuclear fusion. Hence, there is a releases of energy.	½
Alternatively : There is an increase in BE/nucleon both during	
(i) Nuclear fission of heavy nuclei and	½
(ii) Nuclear fussion of light nuclei	½

(a) Write the basic nuclear process involved in the emission of β^+ in a symbolic form, by a radioactive nucleus.

(b) In the reactions given below :

(i) $${}^{11}_{6}\mathrm{C} \rightarrow {}^{z}_{y}\mathrm{B} + x + \nu$$

(ii) $${}^{12}_{6}\mathrm{C} + {}^{12}_{6}\mathrm{C} \rightarrow {}^{20}_{\mathrm{a}}\mathrm{Ne} + {}^{\mathrm{c}}_{\mathrm{b}}\mathrm{He}$$

Find the values of x, y, and z and a, b and c.

CBSE 2016

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CBSE 2016

a. Basic nuclear reaction $P \rightarrow n + e^+ + \nu$	1
b.(i) $x = \beta^+ / {}^{0}_{1}e$, y =5, z =11	1
(ii) a=10, b=2, c=4	1